

Untitled hardware project

Class edge_ai	Feasibility feasible	BoM Readiness Ready	Score 7.7
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Executive Summary

Metric	Value	Unit	Confidence	Notes
Inference throughput	200	TOPS	medium	Accelerator throughput requested in the brief before model, batch-size, latency and thermal derating evidence.
Estimated CAPEX	0	GBP	low	No CAPEX claim admitted yet; supplier-backed BoM evidence is required before costs appear.
Estimated annual OPEX	0	GBP/year	low	No OPEX claim admitted yet because CAPEX has no sourced BoM basis.

Brief Intake Gate

brief_ready_for_architecture

6/6 intake areas pass; 0 review, 0 blocked. Class edge_ai (high); requirements 3; numeric facts 3; requirement trace 3/3 covered; scratch grammar yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_text	pass	1	246 non-whitespace character(s) in the source brief.	none	Brief text is present.
requirement_quantification	pass	1	3 quantified requirement(s), 3 numeric fact(s).	none	Quantified requirements are sufficient for a first architecture pass.
product_class_selection	pass	1	Selected edge_ai with high confidence.	none	Product class selection is strong enough for scratch architecture.
scratch_design_support	pass	1	edge_ai has a deep scratch architecture grammar.	none	Use scratch universal architecture flow for this brief.
requirement_trace_seed	pass	1	3 covered, 0 partial, 0 uncovered requirement trace row(s).	none	Extracted requirements have trace seeds into the architecture.
assumption_boundary	pass	1	1 brief assumption note(s); no-quantified-default path: no.	none	Assumption boundary is explicit enough for first-pass architecture.

Brief Clarification Plan

no_clarification_needed

No clarification questions are needed before scratch architecture generation.

Architecture Admission Gate

architecture_generation_admitted

6/6 admission areas pass; 0 review, 0 blocked. Can use for review: yes; can proceed to BoM: yes. Source scratch_universal_architecture; next action: none.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_intake_clearance	pass	1	6/6 intake areas pass; class edge_ai (high); 3 quantified requirement(s).	none	Brief intake is strong enough to admit scratch architecture generation.
clarification_clearance	pass	1	0 clarification question(s); 0 required, 0 recommended, 0 architecture-blocking.	none	No brief clarification is needed before architecture generation.
class_grammar_clearance	pass	1	Product class edge_ai; scratch universal grammar supported: yes.	none	Route this class through the scratch universal module grammar.
scratch_stage_clearance	pass	1	Architecture stage passed; source scratch_universal_architecture; modules 10; submodules 30; components 120.	none	Architecture was produced by the scratch universal architecture stage.
architecture_readiness_boundary	pass	1	Architecture ready for BoM: yes; blocking issues 0; required links 8/8.	none	Deterministic architecture readiness checks passed; sourcing work may begin only with explicit evidence.
bom_provenance_hold	pass	1	0 admitted sourcing evidence row(s), 0 source-backed BoM claim(s), 0 provenance violation(s).	none	BoM fields remain behind the explicit sourcing-evidence boundary.

Stage Integrity Gate

stage_trace_accepted

5/5 stage-integrity areas pass; 0 review, 0 blocked. Stages 8/8; ordered: yes; architecture source scratch_universal_architecture; admitted priced lines 0; provenance violations 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
stage_sequence	pass	1	8/8 expected compiler stages are present in order.	none	Stage sequence matches the scratch compiler contract.
stage_payloads	pass	1	8/8 stages have metrics; 8/8 have evidence or explicit limitation; 8/8 have limitations.	none	Every stage carries metrics, evidence or explicit limitation, and limitations.
scratch_architecture_source	pass	1	Product class edge_ai; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Architecture came from the scratch universal grammar.
architecture_before_sourcing	pass	1	Architecture ready: yes; readiness stage passed; sourcing stage warning.	none	Sourcing stage is downstream of architecture readiness and has a consistent status.
sourcing_provenance_boundary	pass	1	0 admitted source evidence row(s), 0 admitted priced line(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN and cost fields stay behind admitted sourcing evidence.

Scratch Lineage Gate

scratch_lineage_clean

6/6 lineage areas pass; 0 review, 0 blocked. Architecture source scratch_universal_architecture; design refs 270; source refs 0; forbidden refs 0; forbidden stage mentions 0; chain-v2 benchmark no (prototype internal depth target).

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_source	pass	1	Product class edge_ai; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Design content came from the scratch universal architecture stage.
design_provenance_refs	pass	1	517 provenance ref(s); 270 design/architecture ref(s); 0 invalid kind(s); 0 forbidden ref(s).	none	All design provenance references are in the allowed scratch provenance vocabulary.
bom_candidate_provenance	pass	1	120 candidate BoM line(s); 0 admitted source evidence row(s); 0 unprovenanced supplier/manufacturer/MPN/cost claim(s).	none	Candidate BoM fields remain unpriced or explicitly source-backed.
source_evidence_boundary	pass	1	0 source provenance ref(s); 0 outside admitted evidence boundary; 0 inside architecture content.	none	Source refs appear only in admitted source evidence, source ledger entries or matching source-backed BoM lines.
chain_v2_quarantine	pass	1	0 forbidden provenance ref(s); 0 forbidden stage-trace mention(s).	none	No chain-v2/state.json/modifier lineage appears inside generated design content or stage traces.
benchmark_isolation	pass	1	Benchmark source: prototype internal depth target; chain benchmark used: no; policy: Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.	none	Benchmark does not use chain-v2 input.

Architecture Freeze Gate

architecture_freeze_review_required

2/6 freeze areas pass; 4 review, 0 blocked. Modules/submodules/components 10/30/120. Structurally ready for sourcing yes; independent review accepted no. Module engineering module_engineering_review_required; interfaces interface_review_ready; requirements coverage_review_ready; next action: Treat the architecture as review-ready until module review questions and critical sourcing blocks close..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
admission_and_lineage	pass	1	Architecture admission architecture_generation_admitted; stage integrity stage_trace_accepted; scratch lineage scratch_lineage_clean.	none	Architecture was admitted from the scratch flow with clean stage and lineage evidence.
module_structure	review	0.75	Component allocation allocation_ready; module engineering module_engineering_review_required; 0/10 modules pass; 8 modules still have critical sourcing blocks.	none	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.
interface_contracts	review	1	Interface verification interface_review_ready; 8/8 required interfaces have carriers; 0 accepted.	none	Collect or accept interface-review evidence before calling the architecture fully frozen.
requirement_coverage	review	1	Requirement coverage coverage_review_ready; 3/3 requirements structurally ready or accepted; accepted ratio 0.	none	Requirement coverage is structurally ready; collect accepted reviewer evidence before full freeze.
engineering_review_state	review	0.83	Pre-BoM engineering engineering_review_ready; 5/6 areas pass; calculations needing review 0; assumptions needing review 23.	none	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.
sourcing_boundary	pass	1	120 candidate BoM lines; 0 admitted sourcing evidence rows; 0 provenance violation(s); 48 critical missing source claim(s).	none	Sourcing and BoM claims remain behind explicit admitted source evidence.

Architecture Freeze Closure Plan

architecture_freeze_review_required

4 closure row(s): 4 ready, 0 blocked. Queues S/E/V/R 1/1/2/0; next row freeze-close:1:module_structure. Structurally ready for sourcing yes; independent review accepted no.

#	Area	Queue	Priority	Status	Required Evidence	Acceptance Criteria	Input Artifacts
1	module_structure	sourcing_intake	major	ready	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.	Module engineering verdict is module_engineering_ready. No module has blocked allocation or unresolved critical sourcing blockers. Every module review question is accepted, revised or explicitly deferred.	*.module-engineering-gate.csv; *.component-allocation-gate.csv; *.engineering-review-pack.csv; *.sourcing-intake-template.csv
2	interface_contracts	verification_intake	minor	ready	Collect or accept interface-review evidence before calling the architecture fully frozen.	Interface verification verdict is accepted_interfaces. Every required interface has endpoint carriers and accepted reviewer evidence.	*.interface-contracts.json; *.interface-verification-gate.csv; *.verification-intake-template.csv
3	requirement_coverage	verification_intake	minor	ready	Requirement coverage is structurally ready; collect accepted reviewer evidence before full freeze.	Requirement coverage verdict is accepted_evidence. Every parsed requirement has architecture links and accepted verification evidence.	*.requirement-coverage-gate.csv; *.engineering-assurance-matrix.csv; *.verification-intake-template.csv
4	engineering_review_state	engineering_review	major	ready	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.	Pre-BoM engineering verdict is engineering_accepted. Calculations, assumptions and verification activities are accepted or explicitly deferred.	*.pre-bom-engineering-gate.csv; *.engineering-calculations.csv; *.engineering-assumptions.csv; *.verification-plan.csv

component_candidates_ready_for_sourcing

7/7 candidate areas pass; 0 review, 0 blocked. Candidate lines 120; critical 12/12; worklist rows 120; duplicate groups 0; provenance violations 0; ready for sourcing yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
candidate_presence	pass	1	120 candidate BoM line(s), 120 distinct component identity key(s).	none	Component candidates exist for downstream sourcing review.
candidate_identity	pass	1	120/120 candidate line(s) have non-generic identity, description and role context.	none	Candidate identities are specific enough for first-pass sourcing worklists.
quantity_basis	pass	1	120/120 candidate line(s) have positive quantity and unit.	none	Candidate quantities have positive values and explicit units.
critical_candidate_coverage	pass	1	12/12 class-critical part(s) are represented by candidate lines; 12 critical BoM line(s).	none	Class-critical parts have candidate lines before sourcing begins.
duplicate_identity_review	pass	1	0 duplicate component identity group(s), 0 involving critical lines.	none	No duplicate component identities need sourcing review.
sourcing_worklist_alignment	pass	1	120/120 unpriced candidate line(s) appear in the sourcing worklist.	none	Every unpriced candidate line is visible in the sourcing worklist.
provenance_boundary	pass	1	0 source evidence row(s), 0 source-backed claim(s), 0 provenance violation(s).	none	Candidate sourcing fields remain clean unless admitted evidence supplies them.

Sourcing Authorization Gate

sourcing_authorized

6/6 sourcing-authorization areas pass; 0 review, 0 blocked. Critical intake 12/12; full intake 120/120; admitted evidence 0; rejected evidence 0; provenance violations 0; authorized yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_authorization	pass	1	architecture_generation_admitted; review yes, BoM yes.	none	Architecture admission permits evidence-gated sourcing work.
component_candidate_authorization	pass	1	component_candidates_ready_for_sourcing; 120 candidate line(s), 120 sourcing-worklist row(s), 0 provenance violation(s).	none	Component candidates are concrete enough for sourcing intake.
critical_intake_scope	pass	1	12/12 critical unpriced line(s) appear in the critical sourcing intake template.	none	Critical sourcing intake scope is complete.
full_worklist_scope	pass	1	120/120 total unpriced line(s) appear in the full sourcing intake template.	none	Full sourcing intake scope matches the unpriced worklist.
evidence_admission_boundary	pass	1	0 admitted source evidence row(s), 0 rejected source evidence row(s).	none	No rejected sourcing evidence is waiting for repair.
provenance_boundary	pass	1	0 source-backed BoM claim(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN, lead-time and cost fields remain behind admitted evidence.

BoM Admission Gate

candidate_bom_authorized

Display mode candidate_only. 2/7 BoM-admission areas pass; 5 review, 0 blocked. Critical priced 0/12; source-backed claims 0; provenance violations 0. Candidate BoM yes; priced BoM no; procurement use no.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_authorization	pass	1	sourcing_authorized; authorized yes; 6/6 areas pass.	none	Sourcing is authorized, so the BoM may be shown as a candidate sourcing queue.
candidate_bom_scope	pass	1	120 candidate BoM line(s), 12 critical line(s).	none	Candidate BoM scope exists for sourcing and evidence collection.
source_admission_state	review	0	0 admitted source row(s), 0 rejected source row(s), 0/120 priced line(s).	none	No source rows are admitted; keep this as a candidate-only BoM.
critical_bom_admission	review	0	0/12 critical line(s) have admitted unit costs; 12 critical line(s) remain unpriced.	none	Start with critical sourcing intake before treating any BoM cost as admitted.
source_reference_quality	review	0	no_sourcing_evidence; 0/0 source-quality rows pass; 0 protocol fixture row(s).	none	No source evidence is present; keep the BoM candidate-only.
provenance_boundary	review	0.92	0 source-backed claim(s), 48 critical missing-source claim(s), 0 provenance violation(s).	none	Missing critical source claims are acceptable only in candidate-only mode; collect evidence before priced BoM use.
display_mode	review	0.25	Display mode will be candidate_only.	none	Show candidate BoM as a sourcing queue with supplier, MPN and cost fields empty.

Report Readiness Gate

architecture_review_ready

5/6 sections at or above 8. Mean score 7.67. Architecture ready: yes. Critical unpriced lines: 12. Verification accepted: 0/31; unaccepted: 31. Verification acceptance ratio: 0.

- bom score 1 is below 8.
- 12 critical BoM line(s) lack source-backed cost evidence.
- 8 verification activity/activities remain blocked.
- 0/31 non-sourcing verification activity/activities have accepted reviewer evidence.

Section	Score	Target	B/M/m Issues	Action
executive_summary	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
brief_requirements	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
design_modules	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
bom	1	below target	12/1/0	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates.
feasibility_notes	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
sources_references	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.

Pre-BoM Engineering Gate

engineering_review_ready

5/6 areas pass; 1 review, 0 blocked. Modules/submodules/components: 10/30/120. Calculations within envelope 2/2; calculation blockers 0. Assumption review rows 23; architecture blockers 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	10 modules, 30 submodules and 120 component words pass architecture readiness.	none	Architecture is structurally ready for engineering review.
component_allocation	pass	1	30/30 submodules ready; 12/12 critical parts allocated; 0 duplicate component groups.	none	Component candidates are allocated well enough for engineering review.
interface_verification	pass	1	8/8 required interfaces have endpoint carriers; 0 accepted with evidence.	none	Required interfaces are structurally ready for reviewer evidence.
requirement_coverage	pass	1	3/3 requirements are structurally ready or accepted; accepted evidence ratio 0.	none	Requirements are structurally ready for reviewer evidence.
calculation_envelope	pass	1	2/2 calculations are within envelope; 0 need review; 0 block.	none	Calculation envelope has no deterministic blocker.
engineering_assumptions	review	0.32	11/34 assumptions are brief-supported or model-present; 23 need review/source evidence; 0 block architecture.	none	Review assumptions and keep BoM/source-required assumptions out of trusted claims until evidence is admitted.

BoM Costing Gate

costing_not_started

1/7 costing areas pass; 0 review, 6 blocked. Critical priced lines 0/12; unpriced critical lines 12. Source-backed claims 0; provenance violations 0; sourcing evidence 0/0 production-ready, 0 protocol. Source quality: no_sourcing_evidence (0/0 pass, 0 placeholder URL, 0 quote-anchored). BoM total £0; CAPEX £0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_admission	blocked	0	0/120 BoM lines have admitted pricing; rejected source records 0.	structure_containment_rack_chassis_1u_rack_chassis: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. compute_acceleration_accelerator_module_ai_accelerator_module: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. compute_acceleration_cpu_memory_complex_cpu_module: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. compute_acceleration_cpu_memory_complex_ecc_memory: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. compute_acceleration_cpu_memory_complex_system_motherboard: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 7 more	Admit source-backed supplier, manufacturer, MPN and unit-cost evidence before costing review.
critical_line_pricing	blocked	0	0/12 critical lines have admitted unit costs.	structure_containment_rack_chassis_1u_rack_chassis: 1U rack chassis compute_acceleration_accelerator_module_ai_accelerator_module: AI accelerator module compute_acceleration_cpu_memory_complex_cpu_module: CPU module compute_acceleration_cpu_memory_complex_ecc_memory: ECC memory compute_acceleration_cpu_memory_complex_system_motherboard: System	Price every critical line from admitted source evidence.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				motherboard plus 7 more	
source_backed_provenance	blocked	0.92	0 source-backed claims; 48 critical missing-source claims; 0 provenance violations.	structure_containment_rack_chassis_1_u_rack_chassis/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_rack_chassis_1_u_rack_chassis/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_rack_chassis_1_u_rack_chassis/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_rack_chassis_1_u_rack_chassis/unit_cost_gbp: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. compute_acceleration_accelerator_module_ai_accelerator_module/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 43 more	Keep supplier, manufacturer, MPN, unit cost and lead-time fields empty until source-backed evidence is admitted.
component_identity	pass	1	0 duplicate component identity group(s); 0 include critical lines.	none	Component identities are unique enough for line-level costing.
sourcing_evidence_authenticity	blocked	0	0/0 sourcing evidence rows are production-ready; 0 protocol fixture, 0 review/missing metadata.	none	Add sourcing evidence rows before BoM costing can start.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
source_reference_qualified	blocked	0	0/0 sourcing refs pass quality checks; 0 placeholder URL, 0 quote-anchored, 0 fresh timestamp.	none	Add source references before costing review.
cost_totals	blocked	0	BoM total 0 GBP; CAPEX 0 GBP; annual OPEX 0 GBP.	12 critical line(s) are unpriced, so totals are incomplete.	Do not use cost totals until every critical line has an admitted unit cost.

Evidence Acquisition Plan

48 missing-evidence row(s): 15 sourcing, 33 verification. Ready 45; blocked 0. Blocks BoM 15; blocks publish 48. Next row: acquire:4:assumption:metric:capex_gbp.

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
1	sourcing	blocker	open	readiness:2	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
2	verification	blocker	open	readiness:3	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
3	verification	blocker	open	readiness:4	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
4	sourcing	blocker	ready_for_intake	assumption:metric:capex_gbp	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						sourceGrade; evidence.ref; evidence.quote; retrievedAt		
5	sourcing	blocker	ready_for_intake	assumption:metric:opex_annual_gbp	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
6	sourcing	blocker	ready_for_intake	1u_rack_chassis	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
7	sourcing	blocker	ready_for_intake	ai_accelerator_module	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidence.ref ; evidence.quote; retrievedAt		
8	sourcing	blocker	ready_for_intake	cpu_module	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
9	sourcing	blocker	ready_for_intake	ecc_memory	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
10	sourcing	blocker	ready_for_intake	system_motherboard	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidence.quote; retrievedAt		
11	sourcing	blocker	ready_for_intake	high_speed_nic	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
12	sourcing	blocker	ready_for_intake	nvme_ssd	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
13	sourcing	blocker	ready_for_intake	bmc_management_controller	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						ote; retrievedAt		
14	sourcing	blocker	ready_for_in take	redundant_ power_suppl y	supplier_cat alogue_or_q uote	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes
15	sourcing	blocker	ready_for_in take	fan_wall_as sembly	supplier_cat alogue_or_q uote	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
16	sourcing	blocker	ready_for_intake	heatsink_cold_plate	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
17	sourcing	blocker	ready_for_intake	tpm_security_module	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
18	verification	major	ready_for_intake	design_review:structure_containment	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
19	verification	major	ready_for_intake	calculation:structure_containment	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef;	anonymous reviewer; unchecked model assertion; test-fixture://	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidenceNote; reviewedAt	review; source-evidence claims that belong in sourcing intake	
20	verification	major	ready_for_intake	design_review:compute_acceleration	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
21	verification	major	ready_for_intake	calculation:compute_acceleration	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
22	verification	major	ready_for_intake	design_review:network_io	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
23	verification	major	ready_for_intake	design_review:data_storage	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
							sourcing intake	
24	verification	major	ready_for_intake	design_review:control_communication	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
25	verification	major	ready_for_intake	calculation: control_compute_communication	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
26	verification	major	ready_for_intake	design_review:power_distribution	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
27	verification	major	ready_for_intake	calculation: power_distribution	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
28	verification	major	ready_for_intake	design_review:environmental_interface	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
29	verification	major	ready_for_intake	calculation:environmental_interface	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
30	verification	major	ready_for_intake	design_review:sensing_instrumentation	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
31	verification	major	ready_for_intake	design_review:safety_protection	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
32	verification	major	ready_for_intake	calculation:safety_protection	named_engineering_review	activityId; evidenceKind; reviewerNa	anonymous reviewer; unchecked model	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						me; verdict; evidenceRef; evidenceNote; reviewedAt	assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	
33	verification	major	ready_for_intake	design_review:maintenance_serviceability	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
34	verification	major	ready_for_intake	calculation:maintenance_serviceability	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
35	verification	major	ready_for_intake	interface_review:control_compute_communication:compute_acceleration:pcie_fabric	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
36	verification	major	ready_for_intake	interface_review:power_distribution:compute_acceleration:gpu_power_bus	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNo	anonymous reviewer; unchecked model assertion; test- fixture:// review; source-	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						te; reviewedAt	evidence claims that belong in sourcing intake	
37	verification	major	ready_for_in take	interface_re view:networ k_io:control _compute_c ommunicati on:inference _network	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
38	verification	major	ready_for_in take	interface_re view:data_s torage:contr ol_compute _communic ation:storag e_bus	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
39	verification	major	ready_for_in take	interface_re view:enviro nmental_int erface:com pute_accele ration:therm al_path	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
40	verification	major	ready_for_intake	interface_review:sensing_instrumentation:control_computer_communication:sensor_bus	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
41	verification	major	ready_for_intake	interface_review:safety_protection:power_distribution:hardwired_trip	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
42	verification	major	ready_for_intake	interface_review:structure_containment:environmental_interface:airflow_path	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
43	verification	major	ready_for_intake	compliance_review:IEC-62368-1	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
44	verification	major	ready_for_intake	compliance_review:CE-EMC	named_engineering_review	activityId; evidenceKind; reviewerName	anonymous reviewer; unchecked model	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						me; verdict; evidenceRef ; evidenceNote; reviewedAt	assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	
45	verification	major	ready_for_intake	compliance_review:SECURE-BOOT-REVIEW	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
46	verification	major	ready_for_intake	risk_review:1	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
47	verification	major	ready_for_intake	risk_review:2	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
48	verification	major	ready_for_intake	risk_review:3	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef ; evidenceNo	anonymous reviewer; unchecked model assertion; test- fixture:// review; source-	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						te; reviewedAt	evidence claims that belong in sourcing intake	

Document Trust Gate

evidence_blocked

1/7 trust areas pass; 3 review, 3 blocked. Architecture ready: yes. Claim gate: evidence_blocked. Reviewer evidence 0/31. Source-backed BoM claims 0; critical missing BoM claims 48; provenance violations 0. Evidence authenticity: no_evidence (0/0 production-ready, 0 protocol).

- bom: score 1 below 8
- architecture design has 0 blocked/source-required claim(s) and 40 review-required claim(s).
- engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s).
- bom sourcing has 60 blocked/source-required claim(s) and 108 review-required claim(s).
- compliance risk has 0 blocked/source-required claim(s) and 6 review-required claim(s).
- structure_containment_rack_chassis_1u_rack_chassis/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- structure_containment_rack_chassis_1u_rack_chassis/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- structure_containment_rack_chassis_1u_rack_chassis/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- structure_containment_rack_chassis_1u_rack_chassis/unit_cost_gbp: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- compute_acceleration_accelerator_module_ai_accelerator_module/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- compute_acceleration_accelerator_module_ai_accelerator_module/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- compute_acceleration_accelerator_module_ai_accelerator_module/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	10 modules, 30 submodules and 8 required links pass deterministic architecture gates.	none	No deterministic architecture blocker remains; continue with reviewer evidence before treating it as accepted.
section_scores	blocked	0.83	5/6 scored sections are at or above 8; mean 7.67.	bom: score 1 below 8	Repair below-target sections through the specific score-ledger actions.
claim_evidence	blocked	0.01	3/222 claims pass; 62 blocked/source-required, 157 review-required.	architecture design has 0 blocked/source-required claim(s) and 40 review-required claim(s). engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s). bom sourcing has 60 blocked/source-required claim(s) and 108 review-required claim(s). compliance risk has 0 blocked/source-required claim(s) and 6 review-required claim(s).	Accept, correct or reject generated module and submodule allocation claims through engineering review. Accept, correct or reject calculated metrics and engineering envelopes through reviewer evidence. Admit source-backed supplier, manufacturer, MPN, unit-cost and lead-time evidence for required BoM claims. Accept, correct or reject class-pack regulatory and risk claims through compliance/domain review.
requirement_assurance	review	0	0/3 requirements have accepted assurance; 3 need review, 0 blocked, 0 unlinked.	none	Collect reviewer evidence for requirement assurance rows before publication.
reviewer_evidence	review	0	0/31 non-sourcing verification activities accepted; 0 rejected, 0 deferred, 31 pending/blocked.	none	Collect accepted reviewer evidence for every non-sourcing verification activity.
bom_provenance	blocked	0.92	0 source-backed BoM claims, 48 critical missing-source claims, 0 provenance violations.	structure_containment_rack_chassis_1 u_rack_chassis/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_rack_chassis_1 u_rack_chassis/manufacturer: Admit	Admit source-backed evidence for critical BoM supplier, manufacturer, MPN and cost claims.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_rack_chassis_1 u_rack_chassis/mpn : Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_rack_chassis_1 u_rack_chassis/unit_cost_gbp: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. compute_acceleration_accelerator_module_ai_accelerator_module/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 43 more	

evidence_authenticity	review	0	0/0 evidence row(s) are production-ready; 0 protocol fixture, 0 review/missing metadata.	none	Admit source and reviewer evidence before trusting the document.
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Evidence Authenticity Gate

no_evidence

0/0 evidence rows are production-ready. Protocol fixtures 0; review-required 0; missing metadata 0. External URLs 0; internal references 0; local files 0; unknown references 0.

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
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Evidence Replacement Plan

No evidence replacement rows remain for the current authenticity verdict: no_evidence.

Trust Repair Plan

6 package(s): 6 ready, 0 waiting. Next package: sourcing_intake. Closure rows 58; trust blockers 61. Trust verdict: evidence_blocked.

#	Package	Status	Gate Areas	Closure Rows	Trust Blockers	Required Inputs	Source Artifacts	Exit Criteria
2	Admit Source-Backed BoM Claims	ready	claim_evidence; bom_provenance	15	52	Supplier, manufacturer, MPN, unit cost, lead time, retrieved-at timestamp and source reference for critical BoM lines. Rejected or duplicate component identities resolved before pricing shared items.	*.sourcing-intake-template.csv; *.sourcing-pack.csv; *.bom-provenance-manifest.csv; *.component-identity.csv	BoM provenance has zero critical missing-source claims. Claim Evidence Gate BoM sourcing area is no longer evidence_blocked.
3	Resolve Engineering Review Questions	ready	claim_evidence; requirement_assurance	9	4	Accepted/corrected/rejected answers to review-pack questions. Calculation and assumption decisions tied to reviewer evidence references.	*.engineering-review-pack.csv; *.engineering-assurance-matrix.csv; *.engineering-assumptions.csv; *.engineering-calculations.csv	Requirement assurance rows have accepted or explicitly deferred engineering review evidence. Generated architecture and calculation claims are accepted, corrected or rejected.
4	Admit Reviewer Verification Evidence	ready	reviewer_evidence	33	0	Accepted, rejected or deferred verification evidence for every non-source verification activity.	*.verification-intake-template.csv; *.verification-plan.csv; *.verification-ledger.csv	Every non-source verification activity has accepted reviewer evidence or an explicit defer decision.
5	Validate Evidence Authenticity	ready	evidence_authenticity	0	0	Production supplier/catalogue URLs for BoM source records. Named reviewer evidence references that are not protocol fixtures.	*.evidence-authenticity-gate.csv; *.sourcing-ledger.csv; *.verification-ledger.csv; *.claim-ledger.csv	Evidence Authenticity Gate verdict is production_ready. No protocol_fixture, unknown_reference or missing_metadata evidence

#	Package	Status	Gate Areas	Closure Rows	Trust Blockers	Required Inputs	Source Artifacts	Exit Criteria
								rows remain.
6	Review Compliance And Risk Claims	ready	claim_evidence	0	4	Compliance-domain review of class-pack standards, hazards and mitigation claims.	*.claim-ledger.csv; *.engineering-review-pack.csv	Compliance and risk claims are accepted, corrected or rejected by a domain reviewer.
7	Repair Residual Section Scores	ready	section_scores	1	1	A rerun after evidence intake, plus score-ledger fixes for any section still below target.	*.score-ledger.csv; *.readiness-gate.csv; *.document-trust-gate.csv	All scored sections are at or above target and Document Trust Gate is no longer blocked by scoring.

Claim Ledger

222 claims: 3 brief-supplied, 0 reviewer-accepted, 0 source-backed, 3 calculated needing review, 154 generated needing review, 62 source-required, 0 blocked. Blocking: 62 BoM, 217 publishable.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_required	assumption	AI accelerator module / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	AI accelerator module / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	AI accelerator module / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	AI accelerator module / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	CPU module / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	CPU module / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								before BoM can pass.
bom_source_field	source_required	assumption	CPU module / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	CPU module / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	ECC memory / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	ECC memory / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	ECC memory / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	ECC memory / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								cost evidence before BoM can pass.
bom_source_field	source_required	assumption	System motherboard / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	System motherboard / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	System motherboard / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	System motherboard / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	BMC management controller / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	BMC management controller / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/ma

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	BMC management controller / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	BMC management controller / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	NVMe SSD / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	NVMe SSD / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	NVMe SSD / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	NVMe SSD / unit_cost_gbp	unit_cost_gbp is	none	yes	yes	Admit source-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
			bp	required but not source-backed.				backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Fan wall assembly / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Fan wall assembly / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Fan wall assembly / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Fan wall assembly / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Heatsink cold plate / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_required	assumption	Heatsink cold plate / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Heatsink cold plate / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Heatsink cold plate / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	High-speed NIC / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	High-speed NIC / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	High-speed NIC / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								before BoM can pass.
bom_source_field	source_required	assumption	High-speed NIC / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Redundant power supply / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Redundant power supply / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Redundant power supply / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Redundant power supply / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	TPM security module / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								cost evidence before BoM can pass.
bom_source_field	source_required	assumption	TPM security module / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	TPM security module / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	TPM security module / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	1U rack chassis / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	1U rack chassis / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	1U rack chassis / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/ma

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	1U rack chassis / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
component_candidate	source_required	generated_model	AI accelerator module	AI accelerator module is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.edge_ai	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	CPU module	CPU module is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.edge_ai	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	ECC memory	ECC memory is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.edge_ai	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	System motherboard	System motherboard is an architecture	model:scratch_universal_architecture.edge_ai	yes	yes	Admit source-backed supplier,

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	source_required	generated_model	BMC management controller	-derived component candidate with quantity 1 candidate.				manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
				BMC management controller is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.edge_ai	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	NVMe SSD	NVMe SSD is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.edge_ai	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Fan wall assembly	Fan wall assembly is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.edge_ai	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	source_required	generated_model	Heatsink cold plate	Heatsink cold plate is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.edge_ai	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	High-speed NIC	High-speed NIC is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.edge_ai	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Redundant power supply	Redundant power supply is an architecture-derived component candidate with quantity 2 each.	model:scratch_universal_architecture.edge_ai	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	TPM security module	TPM security module is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.edge_ai	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	1U rack chassis	1U rack chassis is an architecture-derived component candidate with	model:scratch_universal_architecture.edge_ai	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
				quantity 1 candidate.				time evidence through sourcing intake.
headline_metric	source_required	assumption	Estimated CAPEX	Estimated CAPEX = 0 GBP.	formula:capex=sum(bom.totalCostGbp); class_pack:edge_ai.class_pack	yes	yes	Admit source-backed BoM costs before treating this cost metric as a claim.
headline_metric	source_required	assumption	Estimated annual OPEX	Estimated annual OPEX = 0 GBP/year.	formula:opex=capex*0.04	yes	yes	Admit source-backed BoM costs before treating this cost metric as a claim.
component_candidate	generated_needs_review	generated_model	accelerator carrier board	accelerator carrier board is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.edge_ai	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	accelerator retention latch	accelerator retention latch is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.edge_ai	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	PCIe edge connector	PCIe edge connector is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.edge_ai	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	boot flash device	boot flash device is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.edge_ai	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	generated_needs_review	generated_model	clock distribution device	clock distribution device is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.edge_ai	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	high-current VRM module	high-current VRM module is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.edge_ai	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	PCIe retimer set	PCIe retimer set is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.edge_ai	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	power integrity capacitor bank	power integrity capacitor bank is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.edge_ai	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Showing first 70 of 222 claims; export CSV for the full claim ledger.

Claim Evidence Gate

evidence_blocked

3/222 claims pass by being brief-supplied, source-backed or reviewer-accepted. 62 blocked/source-required claims, 157 review-required claims. Claim pass ratio 0.01. Blocking: 62 BoM, 217 publishable.

- architecture design has 0 blocked/source-required claim(s) and 40 review-required claim(s).
- engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s).
- bom sourcing has 60 blocked/source-required claim(s) and 108 review-required claim(s).
- compliance risk has 0 blocked/source-required claim(s) and 6 review-required claim(s).

Area	Verdict	Claims	Passed	Review Req.	Blocked	Source Req.	Blocks BoM	Blocks Publish	Pass Ratio	Required Action
brief_requirements	complete	3	3	0	0	0	0	0	1	No claim-evidence action required for this area.
architecture_design	review	40	0	40	0	0	0	40	0	Accept, correct or reject generated module and submodule allocation claims through engineering review.
engineering_math	blocked	5	0	3	2	2	2	3	0	Accept, correct or reject calculated metrics and engineering envelopes through reviewer evidence.
bom_sourcing	blocked	168	0	108	60	60	60	168	0	Admit source-backed supplier, manufacturer, MPN, unit-cost and lead-time evidence for required BoM claims.
compliance_risk	review	6	0	6	0	0	0	6	0	Accept, correct or reject class-pack regulatory and risk claims through compliance

Area	Verdict	Claims	Passed	Review Req.	Blocked	Source Req.	Blocks BoM	Blocks Publish	Pass Ratio	Required Action
										ce/domain review.

Evidence Gap Register

58 open evidence gaps: 18 blockers, 40 major, 0 minor. Closure paths: 15 sourcing intake, 33 verification intake, 9 engineering review, 0 architecture revision. Blocking rows: architecture 0, BoM 15, publishable 58.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
readiness_blocker	blocker	open	Report readiness gate	bom score 1 is below 8.	score_repair	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	12 critical BoM line(s) lack source-backed cost evidence.	sourcing_intake	bom; publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	8 verification activity/activities remain blocked.	verification_intake	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	0/31 non-sourcing verification activity/activities have accepted reviewer evidence.	verification_intake	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier,

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							manufacturer, MPN and cost evidence for critical BoM lines.
assumption_review	major	needs review	Inference throughput	Inference throughput uses compute_tops with medium confidence.	engineering_review	publishable	Review the formula, input values and operating envelope before treating this metric as a performance claim.
assumption_review	blocker	ready for intake	Estimated CAPEX	Estimated CAPEX uses an implicit model with low confidence.	sourcing_intake	bom; publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	blocker	ready for intake	Estimated annual OPEX	Estimated annual OPEX uses an implicit model with low confidence.	sourcing_intake	bom; publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	major	needs review	Inference Throughput	Inference Throughput currently evaluates as pass against 10 to 2000 TOPS for a rack edge inference appliance before model and latency derating.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	Rack Power Density	Rack Power Density currently evaluates as pass against 100 to 1500 W/U for forced-air edge appliances; higher density needs detailed rack inlet and cooling proof.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	Data/Compute Closure	Data/Compute Closure currently	engineering_review	publishable	Supply an engineering calculation,

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
				evaluates as pass against PCIe fabric, inference network, storage bus and management network present.			reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	Thermal/Power Safety	Thermal/Power Safety currently evaluates as pass against thermal path, airflow path, fan control, AC input, protective earth, thermal alarm and hardwired trip present.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	Secure Management	Secure Management currently evaluates as pass against secure boot chain, management network, service access and TPM present.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	IEC-62368-1	Audio/video, ICT and communication equipment safety is relevant to this product class in UK/EU/global.	engineering_review	publishable	Electrical safety, enclosure, temperature and abnormal-operation evidence.
assumption_review	major	needs review	CE-EMC	Electromagnetic compatibility technical file is relevant to this product class in UK/EU.	engineering_review	publishable	EMC design review, pre-compliance or test evidence for rack appliance.
assumption_review	major	needs review	SECURE-BOOT-REVIEW	Secure boot and fleet-management review is relevant to this product	engineering_review	publishable	TPM, measured boot, firmware signing, recovery and

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
				class in Project security gate.			management-access evidence.
sourcing_evidence	blocker	ready for intake	1U rack chassis	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	AI accelerator module	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	CPU module	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	ECC memory	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	System motherboard	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	High-speed NIC	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	NVMe SSD	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	BMC management controller	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Redundant power supply	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Fan wall assembly	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Heatsink cold plate	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	TPM security module	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
verification_evidence	major	ready for intake	Rack Structure Containment	Review Rack Structure Containment architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Rack Structure Containment	Verify Rack Structure Containment requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Compute Acceleration	Review Compute Acceleration architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Compute Acceleration	Verify Compute Acceleration requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
verification_evidence	major	ready for intake	Network IO	Review Network IO architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Data Storage	Review Data Storage architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Control Compute Communication	Review Control Compute Communication architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Control Compute Communication	Verify Control Compute Communication requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Power Distribution	Review Power Distribution architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Power Distribution	Verify Power Distribution requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Thermal Airflow Interface	Review Thermal Airflow Interface architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Thermal Airflow Interface	Verify Thermal Airflow Interface requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
verification_evidence	major	ready for intake	Telemetry Instrumentation	Review Telemetry Instrumentation architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Security And Safety Protection	Review Security And Safety Protection architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Security And Safety Protection	Verify Security And Safety Protection requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Field Maintenance Serviceability	Review Field Maintenance Serviceability architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Field Maintenance Serviceability	Verify Field Maintenance Serviceability requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Control Compute Communication -> Compute Acceleration	Verify pcie_fabric interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Power Distribution -> Compute Acceleration	Verify gpu_power_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Network IO -> Control Compute Communication	Verify inference_network interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
verification_evidence	major	ready for intake	Data Storage -> Control Compute Communication	Verify storage_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Thermal Airflow Interface -> Compute Acceleration	Verify thermal_path interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Telemetry Instrumentation -> Control Compute Communication	Verify sensor_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Security And Safety Protection -> Power Distribution	Verify hardwired_trip interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Rack Structure Containment -> Thermal Airflow Interface	Verify airflow_path interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for IEC-62368-1	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for CE-EMC	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for SECURE-BOOT-REVIEW	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Accelerator thermal throttling or shutdown under rack inlet conditions	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Insecure firmware or management interface compromise	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Inference outage from PSU, fan or SSD failure	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Closure Plan

4 active phases. Next phase: sourcing_intake. Rows by phase: architecture 0, sourcing 15, engineering 9, verification 33, score repair 1.

#	Phase	Rows	B/M/m	Blocks	Rationale	Exit Criteria	Top Gap IDs
2	Admit Source-Backed BoM Evidence	15	15/0/0	bom; publishable	Critical BoM costs, suppliers, manufacturers and MPNs must be source-backed before cost claims are credible.	Every critical BoM line has admitted supplier, manufacturer, MPN, unit cost and source reference. BoM provenance manifest has zero critical missing-source claims.	assumption:metric:capex_gbp; assumption:metric:opex_annual_gbp; readiness:2; sourcing:1u_rack_chassis; sourcing:ai_accelerator_module; sourcing:bmc_management_controller; sourcing:cpu_module; sourcing:ecc_memory; plus 7 more
3	Complete Engineering Review	9	0/9/0	publishable	Deterministic arithmetic can expose claims, but engineering judgment must review envelopes, assumptions and operating cases.	All calculation and assumption review rows are accepted, revised or explicitly deferred with reviewer rationale.	assumption:compliance:CE-EMC; assumption:compliance:IEC-62368-1; assumption:compliance:SECURE-BOOT-REVIEW; assumption:metric:headline_output; assumption:sanity:edgeai_compute_throughput; assumption:sanity:edgeai_data_compute_closure; assumption:sanity:edgeai_power_density; assumption:sanity:edgeai_secure_management; plus 1 more
4	Accept Reviewer Evidence	33	2/31/0	publishable	Reviewer acceptance is tracked separately from deterministic scoring and must close before publication.	Every non-source verification activity is accepted, or explicitly deferred and removed from publication criteria.	readiness:3; readiness:4; verification:calculation:compute_acceleration; verification:calculation:control_compute_communication; verification:calculation:environmental_inte

#	Phase	Rows	B/M/m	Blocks	Rationale	Exit Criteria	Top Gap IDs
							rface; verification:ca lculatation:main tenance_servi ceability; verification:ca lculatation:powe r_distribution; verification:ca lculatation:safet y_protection; plus 12 more
5	Repair Remaining Section Scores	1	1/0/0	publishable	After evidence intake, rerun scoring to remove residual deterministic section blockers.	Every scored report section is at or above target and readiness verdict is publishable.	readiness:1

Depth Benchmark

prototype internal depth target. Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.

5 meeting benchmark, 2 below benchmark, average comparable ratio 0.97.

Dimension	Scratch	Benchmark	Ratio	Status	Notes	Action
Functional modules	10 modules	10 modules	1	meets	Core architecture breadth.	Add missing functional modules only when the product class genuinely requires them.
Submodules	30 submodules	30 submodules	1	meets	Submodule count is a rough proxy for engineering decomposition depth.	Expand thin modules with concrete sub-functions and carrier interfaces.
Component candidates	120 components	120 components	1	meets	Component-word count is a rough proxy for BoM review surface, not sourcing quality.	Add justified component candidates where submodules still hide assemblies.
Interface link volume	47 links	36 links	1.31	meets	Compares link volume only; chain-v2 grammar links and scratch interface edges are not semantically identical.	Add interface contracts only where they clarify a real energy, material, signal, load or service path.
Admitted priced-line ratio	0 ratio	0.75 ratio	0	below	Prototype target expects most critical lines to become source-backed before publication.	Use sourcing intake to admit source-backed price/manufacturer/MPN evidence; do not copy benchmark estimates.
Sections scoring >=8	5 sections	6 sections	0.83	below	Readiness gate requires every scored section to reach the target.	Clear section-specific readiness actions, especially BoM evidence.
Verification activities	39 activities	24 activities	1.63	meets	Scratch-only evidence workflow depth; not compared to chain-v2 content.	Keep adding verification activities when new requirements, interfaces or standards are introduced.

Brief Requirements

Design a 1U rack-mount edge AI inference appliance with 200 TOPS accelerator throughput, 700 W power budget, GPU module, redundant power supplies, high-speed NIC, NVMe model cache, BMC management, secure boot and front-to-back thermal management.

Requirement	Value	Unit
Inference throughput	200	TOPS
Rack height	1	U
Power budget	700	W

Requirement Traceability

Requirement	Status	Value	Architecture Coverage	Evaluators	Notes
Inference throughput	covered	200 TOPS	Compute Acceleration / Accelerator module; Thermal Airflow Interface / Forced-air cooling; Control Compute Communication / Host control	metric:headline_output; sanity:edgeai_compute_throughput; sanity:edgeai_data_compute_closure; sanity:edgeai_thermal_power_safety	Throughput is traced through accelerator hardware, runtime control and thermal closure. Requirement has architecture coverage plus metric or engineering sanity coverage.
Rack height	covered	1 U	Rack Structure Containment / Rack chassis; Thermal Airflow Interface / Forced-air cooling; Field Maintenance Serviceability / Field-replaceable units	sanity:edgeai_power_density; sanity:edgeai_thermal_power_safety	Rack height is traced through enclosure, airflow and serviceability constraints. Requirement has architecture coverage plus metric or engineering sanity coverage.
Power budget	covered	700 W	Power Distribution / AC input conversion; Thermal Airflow Interface / Forced-air cooling; Security And Safety Protection / Secure boot	sanity:edgeai_power_density; sanity:edgeai_thermal_power_safety	Power budget is traced through PSU conversion, thermal rejection and hardware shutdown protection. Requirement has architecture coverage plus metric or engineering sanity coverage.

Requirement Coverage Gate

coverage_review_ready

3 requirement row(s): 0 accepted with evidence, 3 coverage-ready, 0 needing review, 0 blocked, 0 unlinked. Structural coverage ratio 1; accepted evidence ratio 0.

Requirement	Status	Present Signals	Missing Signals	Modules	Verification Activities	Required Action
Inference throughput	review ready	architecture_link; submodule_link; component_link; calculation_link; review_question; verification_activity	accepted_verification	compute_acceleration; environmental_interface; control_compute_communication	design_review:compute_acceleration; calculation:compute_acceleration; design_review:control_compute_communication; calculation:control_compute_communication; design_review:environmental_interface; calculation:environmental_interface	Collect accepted reviewer evidence; structural requirement coverage is ready for review.
Power budget	review ready	architecture_link; submodule_link; component_link; calculation_link; review_question; verification_activity	accepted_verification	power_distribution; environmental_interface; safety_protection	design_review:power_distribution; calculation:power_distribution; design_review:environmental_interface; calculation:environmental_interface; design_review:safety_protection; calculation:safety_protection	Collect accepted reviewer evidence; structural requirement coverage is ready for review.
Rack height	review ready	architecture_link; submodule_link; component_link; calculation_link; review_question; verification_activity	accepted_verification	structure_containment; environmental_interface; maintenance_serviceability	design_review:structure_containment; calculation:structure_containment; design_review:environmental_interface; calculation:environmental_interface; design_review:maintenance_serviceability; calculation:maintenance_serviceability	Collect accepted reviewer evidence; structural requirement coverage is ready for review.

Engineering Assurance Matrix

3 requirements: 0 accepted, 0 ready, 3 needing review, 0 blocked, 0 unlinked. 3 have calculation links, 3 have review questions, 0 have accepted verification evidence.

Requirement	Overall	Architecture	Modules	Calculations	Review Questions	Verification	Blockers	Next Action
Inference throughput	needs review	covered	Compute Acceleration; Thermal Airflow Interface; Control Compute Communication	edge_ai_top_s_per_watt	assumption_resolution: component: compute_acceleration_accelerator_module_ai_accelerator_module; assumption_resolution: component: compute_acceleration_cpu_memory_complex_cpu_module; assumption_resolution: component: compute_acceleration_cpu_memory_complex_ecc_memory; assumption_resolution: component: compute_acceleration_cpu_memory_complex_system_motherboard; assumption_resolution: component: control_compute_communication_host_control_bmc_management_controller; plus 27 more	pending; pending; pending; pending; pending	none	Complete engineering review questions, calculations or deferred verification evidence.
Power budget	needs review	covered	Power Distribution; Thermal Airflow Interface; Security And Safety Protection	edge_ai_top_s_per_watt; edge_ai_power_density_w_per_u	assumption_resolution: component: environmental_interface_forced_air_cooling_fan_wall_assembly; assumption_resolution: component: environmental_interface_heat_transfer_heatsink_cold_plate; assumption	pending; pending; pending; pending; pending	none	Complete engineering review questions, calculations or deferred verification evidence.

Requirement	Overall	Architecture	Modules	Calculations	Review Questions	Verification	Blockers	Next Action
					_resolution: component: power_distribution_ac_input_conversion_redundant_power_supply; assumption _resolution: component: safety_protection_security_boot_tpm_security_module; assumption _resolution: sanity:edge_ai_compute_throughput; plus 21 more			
Rack height	needs review	covered	Rack Structure Containment; Thermal Airflow Interface; Field Maintenance Serviceability	edge_ai_power_density_w_per_u	assumption _resolution: component: environment al_interface _forced_air_cooling_fan_wall_assembly; assumption _resolution: component: environment al_interface _heat_transfer_heatsink_cold_plate; assumption _resolution: component: structure_containment _rack_chassis_1u_rack_chassis; assumption _resolution: sanity:edge_ai_compute_throughput; assumption _resolution: sanity:edge_ai_data_compute_closure; plus 18 more	pending; pending; pending; pending	none	Complete engineering review questions, calculations or deferred verification evidence.

Compiler Stage Trace

Stage	Status	Summary	Metrics	Evidence	Limitations
Brief Parsing	passed	Extracted quantified requirements from the user brief before design generation.	requirement_count: 3; numeric_fact_count : 3; requirement_trace_covered: 3; requirement_trace_partial: 0; requirement_trace_uncovered: 0	compute_tops: 200 TOPS; rack_units: 1 U; power_budget_w: 700 W; compute_tops: covered; rack_units: covered; power_budget_w: covered	Parsing is deterministic and currently recognises only a small set of hardware requirement patterns.
Product-Class Selection	passed	Selected edge_ai with high confidence from brief keywords.	selected_class: edge_ai; confidence: high; keyword_scores: edge_ai:7	edge_ai: 7	Classifier is keyword based; ambiguous multi-domain briefs need an LLM or richer taxonomy pass.
Universal Module Architecture	passed	Built a 10-module edge_ai architecture from the scratch universal architecture grammar.	module_count: 10; architecture_source : scratch_universal_architecture	Rack Structure Containment; Compute Acceleration; Network IO; Data Storage; Control Compute Communication; Power Distribution; plus 4 more	The scratch grammar is hand-authored and deterministic; it still needs independent engineering review.
Submodule Expansion	passed	Expanded each module into engineering submodules with local purpose, interfaces and component candidates.	submodule_count: 30; average_submodules_per_module: 3	structure_containment.rack_chassis; structure_containment.accelerator_mounting; structure_containment.service_labelling; compute_acceleration.accelerator_module; compute_acceleration.cpu_memory_complex; compute_acceleration.signal_power_integrity; plus 24 more	Submodule completeness is judged by deterministic coverage checks, not by a physics simulator.
Interface Graph	passed	Checked class-required module-to-module interfaces before allowing BoM review.	required_links: 8; present_links: 8; missing_links: 0; graph_nodes: 40; graph_edges: 77; shared_interface_edges: 39; missing_required_graph_edges: 0	control_compute_communication -> compute_acceleration via pcie_fabric: present; power_distribution -> compute_acceleration via gpu_power_bus: present; network_io -> control_compute_communication via inference_network:	Interfaces prove allocation coverage only; they do not yet validate ratings, thermal margins or dynamic behaviour.

Stage	Status	Summary	Metrics	Evidence	Limitations
				present; data_storage -> control_compute_c ommunication via storage_bus: present; environmental_inter face -> compute_accelerati on via thermal_path: present; sensing_instrument ation -> control_compute_c ommunication via sensor_bus: present; plus 2 more	
Component Candidates	passed	Converted allocated component words into a candidate BoM without supplier prices.	candidate_lines: 120; critical_candidate_li nes: 12; admitted_price_line s: 0	1u_rack_chassis: 1U rack chassis; slide_rail_kit: slide rail kit; internal_divider_pan el: internal divider panel; front_bezel_assemb ly: front bezel assembly; accelerator_retentio n_bracket: accelerator retention bracket; pcie_riser_support: PCIe riser support; plus 24 more	Showing first 30 of 120 component candidates in the stage trace.
Architecture Readiness Gate	passed	Architecture validators found no blocker or major issue, so candidate BoM review can start.	ready_for_bom: true; blocking_issue_cou nt: 0; sanity_pass: 5; sanity_warn: 0; sanity_fail: 0	pass Inference Throughput: 200 TOPS; pass Rack Power Density: 700 W/U; pass Data/Compute Closure: present; pass Thermal/Power Safety: present; pass Secure Management: present	Passing this gate means deterministic architecture checks passed; it is not a certified engineering sign- off.
Sourcing And BoM Admission	warning	No supplier, manufacturer, part number or cost claims have been admitted because no source-backed evidence was provided.	admission_status: not_started; candidate_lines: 120; admitted_lines: 0; unpriced_lines: 120; unpriced_critical_lin es: 12; rejected_source_re cords: 0	none	BoM cost remains intentionally blocked until 12 critical candidate line(s) have explicit source evidence.

Engineering Architecture Review

Ready for BoM review

10 modules, 30 sub-modules, 120 allocated component words.

From	To	Interface	Status	Engineering Reason
control_compute_communication	compute_acceleration	pcie_fabric	OK	Host control must enumerate and command the accelerator over a declared PCIe fabric.
power_distribution	compute_acceleration	gpu_power_bus	OK	Accelerator power must be isolated from lower-power appliance rails.
network_io	control_compute_communication	inference_network	OK	Inference traffic must reach host software and runtime orchestration.
data_storage	control_compute_communication	storage_bus	OK	Model cache and logs must be visible to host software.
environmental_interface	compute_acceleration	thermal_path	OK	Accelerator and CPU heat must couple into the cooling path.
sensing_instrumentation	control_compute_communication	sensor_bus	OK	Thermal and power telemetry must feed management control.
safety_protection	power_distribution	hardwired_trip	OK	Thermal and electrical safety faults must force power shutdown.
structure_containment	environmental_interface	airflow_path	OK	The rack chassis must preserve the front-to-back airflow path used by cooling.

Interface Graph Summary

Module Nodes	Submodule Nodes	Shared Interface Edges	Required Edges	Missing Required Edges
10	30	39	8	0

Interface Contract Matrix

8/8 required contracts present. 16 shared interface families, 2 local-only declarations.

Interface	From Module	From Carrier	To Module	To Carrier	Status	Reason	Notes
pcie_fabric	Control Compute Communication	Host control; Runtime orchestration	Compute Acceleration	Accelerator module; CPU and memory complex; Signal and power integrity	present	Host control must enumerate and command the accelerator over a declared PCIe fabric.	Both endpoints expose the required interface through at least one submodule carrier.
gpu_power_bus	Power Distribution	DC rail distribution	Compute Acceleration	Accelerator module; Signal and power integrity	present	Accelerator power must be isolated from lower- power appliance rails.	Both endpoints expose the required interface through at least one submodule carrier.
inference_network	Network IO	High-speed network; Timing and ingest; IO isolation and routing	Control Compute Communication	Runtime orchestration	present	Inference traffic must reach host software and runtime orchestration.	Both endpoints expose the required interface through at least one submodule carrier.
storage_bus	Data Storage	NVMe storage; Model cache and recovery; Log and evidence buffer	Control Compute Communication	Host control; Runtime orchestration	present	Model cache and logs must be visible to host software.	Both endpoints expose the required interface through at least one submodule carrier.
thermal_path	Thermal Airflow Interface	Heat transfer path	Compute Acceleration	Accelerator module; Signal and power integrity	present	Accelerator and CPU heat must couple into the cooling path.	Both endpoints expose the required interface through at least one submodule carrier.
sensor_bus	Telemetry Instrumentation	Thermal sensors; Power and fan telemetry; Front panel status	Control Compute Communication	Host control	present	Thermal and power telemetry must feed management control.	Both endpoints expose the required interface through at least one submodule carrier.
hardwired_trip	Security And Safety Protection	Thermal shutdown; Electrical safety	Power Distribution	Hold-up and protection	present	Thermal and electrical safety faults must force	Both endpoints expose the required interface

Interface	From Module	From Carrier	To Module	To Carrier	Status	Reason	Notes
						power shutdown.	through at least one submodule carrier.
airflow_path	Rack Structure Containment	Rack chassis; Accelerator mounting	Thermal Airflow Interface	Forced-air cooling; Heat transfer path; Inlet and outlet management	present	The rack chassis must preserve the front-to-back airflow path used by cooling.	Both endpoints expose the required interface through at least one submodule carrier.

Local-only declarations to review later: ac_input_bus (Power Distribution); memory_bus (Compute Acceleration).

Interface Verification Gate

interface_review_ready

8 required interface row(s): 0 accepted, 8 review-ready, 0 pending evidence, 0 deferred, 0 blocked. Structural pass ratio 1; accepted evidence ratio 0.

Interface	From	To	Carrier Status	Verification Activity	Ledger	Status	Required Action
pcie_fabric	Control Compute Communication	Compute Acceleration	both_endpoints_carried	interface_review:control_compute_communication:compute_acceleration:pcie_fabric	pending	review_ready	Collect named interface-review evidence through verification intake.
gpu_power_bus	Power Distribution	Compute Acceleration	both_endpoints_carried	interface_review:power_distribution:compute_acceleration:gpu_power_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
inference_network	Network IO	Control Compute Communication	both_endpoints_carried	interface_review:network_io:control_communication:inference_network	pending	review_ready	Collect named interface-review evidence through verification intake.
storage_bus	Data Storage	Control Compute Communication	both_endpoints_carried	interface_review:data_storage:control_compute_communication:storage_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
thermal_path	Thermal Airflow Interface	Compute Acceleration	both_endpoints_carried	interface_review:environmental_interface:compute_acceleration:thermal_path	pending	review_ready	Collect named interface-review evidence through verification intake.
sensor_bus	Telemetry Instrumentation	Control Compute Communication	both_endpoints_carried	interface_review:sensing_instrumentation:control_compute_communication:sensor_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
hardwired_trip	Security And Safety Protection	Power Distribution	both_endpoints_carried	interface_review:safety_protection:power_distribution:hardwired_trip	pending	review_ready	Collect named interface-review evidence through verification intake.
airflow_path	Rack Structure Containment	Thermal Airflow Interface	both_endpoints_carried	interface_review:structure_containment:environmental_interface:airflow_path	pending	review_ready	Collect named interface-review evidence through verification intake.

Component Allocation Gate

allocation_ready

30/30 submodules ready; 0 review, 0 blocked. Component words 120; critical parts 12/12; duplicate groups 0. Allocation ratio 1; critical allocation ratio 1.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Rack Structure Containment	Rack chassis	ready	4	rack_mount; airflow_path; service_access; protective_earth	1u_rack_chassis	none	Submodule has component candidates and no allocation blocker.
Rack Structure Containment	Accelerator mounting	ready	4	rack_mount; airflow_path; service_access	none	none	Submodule has component candidates and no allocation blocker.
Rack Structure Containment	Service labelling	ready	4	service_access; rack_mount	none	none	Submodule has component candidates and no allocation blocker.
Compute Acceleration	Accelerator module	ready	4	pcie_fabric; gpu_power_buses; thermal_path; sensor_bus	ai_accelerator_module	none	Submodule has component candidates and no allocation blocker.
Compute Acceleration	CPU and memory complex	ready	4	memory_bus; pcie_fabric; low_voltage_rail; sensor_bus	cpu_module; ecc_memory; system_motherboard	none	Submodule has component candidates and no allocation blocker.
Compute Acceleration	Signal and power integrity	ready	4	pcie_fabric; gpu_power_buses; thermal_path	none	none	Submodule has component candidates and no allocation blocker.
Network IO	High-speed network	ready	4	inference_network; pcie_fabric; service_access	high_speed_network	none	Submodule has component candidates and no allocation blocker.
Network IO	Timing and ingest	ready	4	inference_network; management_network; service_access	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Network IO	IO isolation and routing	ready	4	inference_net work; service_access; protective_earth	none	none	Submodule has component candidates and no allocation blocker.
Data Storage	NVMe storage	ready	4	storage_bus; pcie_fabric; service_access	nvme_ssd	none	Submodule has component candidates and no allocation blocker.
Data Storage	Model cache and recovery	ready	4	storage_bus; secure_boot_chain; management_network	none	none	Submodule has component candidates and no allocation blocker.
Data Storage	Log and evidence buffer	ready	4	storage_bus; management_network; service_access	none	none	Submodule has component candidates and no allocation blocker.
Control Compute Communication	Host control	ready	4	pcie_fabric; storage_bus; management_network; sensor_bus	bmc_management_controller	none	Submodule has component candidates and no allocation blocker.
Control Compute Communication	Runtime orchestration	ready	4	pcie_fabric; storage_bus; inference_net work; secure_boot_chain	none	none	Submodule has component candidates and no allocation blocker.
Control Compute Communication	Remote management	ready	4	management_network; service_access; secure_boot_chain	none	none	Submodule has component candidates and no allocation blocker.
Power Distribution	AC input conversion	ready	4	ac_input_bus; protective_earth; sensor_bus	redundant_power_supply	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Power Distribution	DC rail distribution	ready	4	low_voltage_rail; gpu_power_buses; service_access	none	none	Submodule has component candidates and no allocation blocker.
Power Distribution	Hold-up and protection	ready	4	low_voltage_rail; hardwired_trip; sensor_bus	none	none	Submodule has component candidates and no allocation blocker.
Thermal Airflow Interface	Forced-air cooling	ready	4	airflow_path; fan_control_buses; sensor_bus	fan_wall_assembly	none	Submodule has component candidates and no allocation blocker.
Thermal Airflow Interface	Heat transfer path	ready	4	thermal_path; airflow_path; service_access	heatsink_cold_plate	none	Submodule has component candidates and no allocation blocker.
Thermal Airflow Interface	Inlet and outlet management	ready	4	airflow_path; service_access	none	none	Submodule has component candidates and no allocation blocker.
Telemetry Instrumentation	Thermal sensors	ready	4	sensor_bus; thermal_alarm_bus	none	none	Submodule has component candidates and no allocation blocker.
Telemetry Instrumentation	Power and fan telemetry	ready	4	sensor_bus; fan_control_buses; management_network	none	none	Submodule has component candidates and no allocation blocker.
Telemetry Instrumentation	Front panel status	ready	4	sensor_bus; management_network; service_access	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Security And Safety Protection	Secure boot	ready	4	secure_boot_chain; management_network; service_accesses	tpm_security_module	none	Submodule has component candidates and no allocation blocker.
Security And Safety Protection	Thermal shutdown	ready	4	thermal_alarm_bus; hardwired_trip; fan_control_bus	none	none	Submodule has component candidates and no allocation blocker.
Security And Safety Protection	Electrical safety	ready	4	protective_earth; service_accesses; hardwired_trip	none	none	Submodule has component candidates and no allocation blocker.
Field Maintenance Serviceability	Field-replaceable units	ready	4	service_accesses; airflow_path	none	none	Submodule has component candidates and no allocation blocker.
Field Maintenance Serviceability	Rack installation	ready	4	rack_mount; service_accesses	none	none	Submodule has component candidates and no allocation blocker.
Field Maintenance Serviceability	Service workflow	ready	4	service_accesses; management_network; secure_boot_chain	none	none	Submodule has component candidates and no allocation blocker.

Submodule Engineering Gate

submodule_engineering_review_required

0/30 submodules pass; 30 review, 0 blocked. Component words 120; linked requirements 9; carrier contracts 32/32; local-only interfaces 58. Review questions accepted 0/92; verification accepted 0/79; critical unpriced lines 12.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Rack Structure Containment	Rack chassis	review	4	4	1	1/1	0/3	0/4	1	0.6667	Review 3 local-only interface declaration(s): rack_mount, service_access, protective_earth. 3 engineering review question(s) await acceptance. 3 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.

Rack Structure Containment	Accelerator mounting	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): rack_mount, service_access. 3 engineering review question(s) await
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Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											acceptance. 2 verification activity row(s) await accepted evidence.
Rack Structure Containment	Service labelling	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): service_access, rack_mount. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Compute Acceleration	Accelerator module	review	4	4	1	3/3	0/5	0/6	1	0.6667	Review 1 local-only interface declaration(s): sensor_bus. 5 engineering review question(s) await acceptance. 5 verification activity row(s) await acceptance.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
d evidence. 1 critical component line(s) still need source-backed cost/manufactur er/MPN evidence.											
Compute Acceleration	CPU and memory complex	review	4	4	0	1/1	0/3	0/3	3	0.6667	Review 3 local-only interface declaration(s): memory_bus, low_voltage_rail, sensor_bus. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence. 3 critical component line(s) still need source-backed cost/manufactur er/MPN evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Compute Acceleration	Signal and power integrity	review	4	3	0	3/3	0/5	0/4	0	0.6667	5 engineering review question(s) await acceptance. 4 verification activity row(s) await accepted evidence.
Network IO	High-speed network	review	4	3	0	1/1	0/3	0/3	1	0.6667	Review 2 local-only interface declaration(s): pcie_fabric, service_access. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Network IO	Timing and ingest	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): management_network, service_access. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Network IO	IO isolation and routing	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): service_access, protective_earth. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Data Storage	NVMe storage	review	4	3	0	1/1	0/3	0/3	1	0.6667	Review 2 local-only interface declaration(s): pcie_fabric, service_access. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Data Storage	Model cache and recovery	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): secure_boot_chain, management_network. 3 engineering review question(s) await

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											acceptance. 2 verification activity row(s) await accepted evidence.
Data Storage	Log and evidence buffer	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): management_network, service_access. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Control Computer Communication	Host control	review	4	4	1	3/3	0/5	0/6	1	0.6667	Review 1 local-only interface declaration(s): management_network. 5 engineering review question(s) await acceptance. 5 verification

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Control Compute Communication	Runtime orchestration	review	4	4	0	3/3	0/5	0/4	0	0.6667	Review 1 local-only interface declaration(s): secure_boot_chain. 5 engineering review question(s) await acceptance. 4 verification activity row(s) await accepted evidence.
Control Compute Communication	Remote management	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interfaces declaration(s): management_network, service_access, secure_boot_chain. 2

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Power Distribution	AC input conversion	review	4	3	1	0/0	0/2	0/3	1	0.6667	Review 3 local-only interface declaration(s): ac_input_bus, protective_earth, sensor_bus. 2 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Power Distribution	DC rail distribution	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): low_voltage_rail, service_access. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Power Distribution	Hold-up and protection	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): low_voltage_rail, sensor_bus. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Thermal Airflow Interface	Forced-air cooling	review	4	3	3	1/1	0/3	0/4	1	0.6667	Review 2 local-only interface declaration(s): fan_control_bus, sensor_bus. 3 engineering review question(s) await acceptance. 3 verification activity row(s) await acceptance. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Thermal Airflow Interface	Heat transfer path	review	4	3	0	2/2	0/4	0/4	1	0.6667	Review 1 local-only interface declaration(s): service_access. 4 engineering review question(s) await acceptance. 3 verification

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufactur er/MPN evidence.
Thermal Airflow Interface	Inlet and outlet management	review	4	2	0	1/1	0/3	0/2	0	0.6667	Review 1 local-only interface declaration(s): service_access. 3 engineer review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Telemetry Instrumentation	Thermal sensors	review	4	2	0	1/1	0/3	0/2	0	0.6667	Review 1 local-only interface declaration(s): thermal_alarm_bus. 3 engineer review question(s)

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											await acceptance. 2 verification activity row(s) await accepted evidence.
Telemetry Instrumentation	Power and fan telemetry	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): fan_control_bus, management_network. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Telemetry Instrumentation	Front panel status	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): management_network, service_access. 3 engineering review question(s) await

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											acceptance. 2 verification activity row(s) await accepted evidence.
Security And Safety Protection	Secure boot	review	4	3	1	0/0	0/2	0/3	1	0.6667	Review 3 local-only interface declaration(s): secure_boot_chain, management_network, service_access. 2 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Security And Safety Protection	Thermal shutdown	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): thermal_alarm_bus, fan_control_bus. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Security And Safety Protection	Electrical safety	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): protective_earth, service_access. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Field Maintenance Serviceability	Field-replaceable units	review	4	2	1	0/0	0/2	0/2	0	0.6667	Review 2 local-only interfaces declaration(s): service_access, airflow_path. 2 engineering review question(s) await acceptance. 2 verification activity row(s) await acceptance evidence.
Field Maintenance Serviceability	Rack installation	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interfaces declaration(s): rack_mount, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await acceptance evidence.
Field Maintenance	Service workflow	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interfaces

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Serviceability											e declarat ion(s): service_ access, manage ment_n etwork, secure_ boot_ch ain. 2 enginee ring review questio n(s) await accepta nce. 1 verificat ion activity row(s) await accepte d evidenc e.

Module Engineering Gate

module_engineering_review_required

0/10 modules pass; 10 review, 0 blocked. Submodules/components: 30/120. Interface carriers 16/16; review questions accepted 0/96; critical unpriced lines 12 across 8 module(s).

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
Rack Structure Containme nt	review	3	12	1	1/1	0/9	1	0.86	1 critical BoM line(s) need source- backed cost evidence. 9 engineerin g review question(s) await acceptanc e. 4 assumptio n row(s) need review or source evidence.
Compute Accelerati on	review	3	12	1	3/3	0/15	4	0.86	4 critical BoM line(s) need source- backed cost evidence. 15 engineerin g review question(s) await acceptanc e. 11 assumptio n row(s) need review or source evidence.
Network IO	review	3	12	0	1/1	0/6	1	0.71	1 critical BoM line(s) need source- backed cost evidence. 6 engineerin g review question(s) await acceptanc e. 3 assumptio n row(s) need review or

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
									source evidence.
Data Storage	review	3	12	0	1/1	0/6	1	0.71	1 critical BoM line(s) need source- backed cost evidence. 6 engineerin g review question(s) await acceptanc e. 3 assumptio n row(s) need review or source evidence.
Control Compute Communic ation	review	3	12	1	4/4	0/13	1	0.86	1 critical BoM line(s) need source- backed cost evidence. 13 engineerin g review question(s) await acceptanc e. 8 assumptio n row(s) need review or source evidence.
Power Distributio n	review	3	12	1	2/2	0/11	1	0.86	1 critical BoM line(s) need source- backed cost evidence. 11 engineerin g review question(s) await acceptanc e. 6 assumptio n row(s) need

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
									review or source evidence.
Thermal Airflow Interface	review	3	12	3	2/2	0/14	2	0.86	2 critical BoM line(s) need source-backed cost evidence. 14 engineering review question(s) await acceptance. 9 assumption row(s) need review or source evidence.
Telemetry Instrumentation	review	3	12	0	1/1	0/5	0	0.71	5 engineering review question(s) await acceptance. 1 assumption row(s) need review or source evidence.
Security And Safety Protection	review	3	12	1	1/1	0/10	1	0.86	1 critical BoM line(s) need source-backed cost evidence. 10 engineering review question(s) await acceptance. 4 assumption row(s) need review or source evidence.
Field Maintenance	review	3	12	1	0/0	0/7	0	0.86	7 engineering review

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
Serviceabi lity									question(s)) await acceptanc e. 3 assumptio n row(s) need review or source evidence.

Module Review Roll-Up

2 ready, 8 sourcing-blocked, 0 needing attention.

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
Rack Structure Containment	sourcing	3	12	1/1	rack_units	1u_rack_chassis	1 critical line(s) need source-backed evidence before BoM pass.
Compute Acceleration	sourcing	3	12	3/3	compute_tops	ai_accelerator_module; cpu_module; ecc_memory; system_motherboard	4 critical line(s) need source-backed evidence before BoM pass.
Network IO	sourcing	3	12	1/1	none	high_speed_nic	1 critical line(s) need source-backed evidence before BoM pass.
Data Storage	sourcing	3	12	1/1	none	nvme_ssd	1 critical line(s) need source-backed evidence before BoM pass.
Control Compute Communication	sourcing	3	12	4/4	compute_tops	bmc_management_controller	1 critical line(s) need source-backed evidence before BoM pass.
Power Distribution	sourcing	3	12	2/2	power_budget_w	redundant_power_supply	1 critical line(s) need source-backed evidence before BoM pass.
Thermal Airflow Interface	sourcing	3	12	2/2	compute_tops; rack_units; power_budget_w	fan_wall_assembly; heatsink_cold_plate	2 critical line(s) need source-backed evidence before BoM pass.
Telemetry Instrumentation	ready	3	12	1/1	none	none	No critical blockers; 12 candidate line(s) remain for later sourcing.

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
Security And Safety Protection	sourcing	3	12	1/1	power_budget_w	tpm_security_module	1 critical line(s) need source-backed evidence before BoM pass.
Field Maintenance Serviceability	ready	3	12	0/0	rack_units	none	No critical blockers; 12 candidate line(s) remain for later sourcing.

Engineering Review Pack

81 reviewer questions: 0 accepted, 50 ready, 31 needing review, 0 blocked. Coverage: 10 modules, 30 submodules, 8 interfaces, 2 calculations, 31 assumptions.

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
assumption_resolution	major	needs review	CE-EMC	Can the CE-EMC assumption be accepted for architecture review, or must it be replaced with evidence?	EMC design review, pre-compliance or test evidence for rack appliance.	none	none	none	none
assumption_resolution	major	needs review	IEC-62368-1	Can the IEC-62368-1 assumption be accepted for architecture review, or must it be replaced with evidence?	Electrical safety, enclosure, temperature and abnormal-operation evidence.	none	none	none	none
assumption_resolution	major	needs review	SECURE-BOOT-REVIEW	Can the SECURE-BOOT-REVIEW assumption be accepted for architecture review, or must it be replaced with evidence?	TPM, measured boot, firmware signing, recovery and management-access evidence.	none	none	none	none
assumption_resolution	major	needs review	AI accelerator module	Can the AI accelerator module assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	compute_acceleration	none	none	none
assumption_resolution	major	needs review	CPU module	Can the CPU module assumption	Admit supplier, manufacturer, MPN	compute_acceleration	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				Can the assumption be accepted for architecture review, or must it be replaced with evidence?	where applicable, unit cost, lead time and evidence reference through sourcing intake.				
assumption_resolution	major	needs review	ECC memory	Can the ECC memory assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	compute_acceleration	none	none	none
assumption_resolution	major	needs review	System motherboard	Can the System motherboard assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	compute_acceleration	none	none	none
assumption_resolution	major	needs review	BMC management controller	Can the BMC management controller assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	control_compute_communication	none	none	none
assumption_resolution	major	needs review	NVMe SSD	Can the NVMe SSD assumption be accepted for	Admit supplier, manufacturer, MPN where applicable,	data_storage	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				architecture review, or must it be replaced with evidence?	unit cost, lead time and evidence reference through sourcing intake.				
assumption_resolution	major	needs review	Fan wall assembly	Can the Fan wall assembly assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	environmental_interface	none	none	none
assumption_resolution	major	needs review	Heatsink cold plate	Can the Heatsink cold plate assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	environmental_interface	none	none	none
assumption_resolution	major	needs review	High-speed NIC	Can the High-speed NIC assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	network_io	none	none	none
assumption_resolution	major	needs review	Redundant power supply	Can the Redundant power supply assumption be accepted for architecture review, or must it	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference	power_distribution	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				be replaced with evidence?	through sourcing intake.				
assumption_resolution	major	needs review	TPM security module	Can the TPM security module assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	safety_protection	none	none	none
assumption_resolution	major	needs review	1U rack chassis	Can the 1U rack chassis assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	structure_containment	none	none	none
assumption_resolution	major	needs review	control_compute_communication -> compute_acceleration	Can the control_compute_communication -> compute_acceleration assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	pcie_fabric	none	none
assumption_resolution	major	needs review	data_storage -> control_compute_communication	Can the data_storage -> control_compute_communication assumption be	Review the interface carrier, direction, capacity, safety margin and failure	none	storage_buses	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				accepted for architecture review, or must it be replaced with evidence?	mode before detailed design or sourcing.				
assumption_resolution	major	needs review	environmental_interface -> compute_acceleration	Can the environmental_interface -> compute_acceleration assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	thermal_path	none	none
assumption_resolution	major	needs review	network_io -> control_compute_communication	Can the network_io -> control_compute_communication assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	inference_network	none	none
assumption_resolution	major	needs review	power_distribution -> compute_acceleration	Can the power_distribution -> compute_acceleration assumption be accepted for architecture review, or must it be replaced	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	gpu_power_bus	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				with evidence?					
assumption_resolution	major	needs review	safety_protection -> power_distribution	Can the safety_protection -> power_distribution assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	hardwired_trip	none	none
assumption_resolution	major	needs review	sensing_instrumentation -> control_compute_communication	Can the sensing_instrumentation -> control_compute_communication assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	sensor_buses	none	none
assumption_resolution	major	needs review	structure_containment -> environmental_interface	Can the structure_containment -> environmental_interface assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	airflow_path	none	none
assumption_resolution	major	needs review	Estimated CAPEX	Can the Estimated CAPEX assumption be accepted	Admit source-backed BoM pricing before	none	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				for architectu re review, or must it be replaced with evidence?	treating this cost metric as a claim.				
assumptio n_resoluti on	major	needs review	Inference throughput	Can the Inference throughput assumption be accepted for architectu re review, or must it be replaced with evidence?	Review the formula, input values and operating envelope before treating this metric as a performan ce claim.	none	none	none	none
assumptio n_resoluti on	major	needs review	Estimated annual OPEX	Can the Estimated annual OPEX assumption be accepted for architectu re review, or must it be replaced with evidence?	Admit source-backed BoM pricing before treating this cost metric as a claim.	none	none	none	none
assumptio n_resoluti on	major	needs review	Inference Throughp ut	Can the Inference Throughp ut assumption be accepted for architectu re review, or must it be replaced with evidence?	Supply an engineerin g calculation , reviewer note, test plan or datasheet -backed rationale for this envelope.	compute_ accelerati on; environme ntal_interf ace; control_co mpute_co mmunicati on	none	none	none
assumptio n_resoluti on	major	needs review	Data/Com pute Closure	Can the Data/Com pute Closure assumption be accepted for	Supply an engineerin g calculation , reviewer note, test plan or datasheet	compute_ accelerati on; environme ntal_interf ace; control_co mpute_co	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				architectu re review, or must it be replaced with evidence?	-backed rationale for this envelope.	mmunicati on			
assumptio n_resoluti on	major	needs review	Rack Power Density	Can the Rack Power Density assumption be accepted for architectu re review, or must it be replaced with evidence?	Supply an engineerin g calculation , reviewer note, test plan or datasheet -backed rationale for this envelope.	structure_ containme nt; environme ntal_interf ace; maintenanc e_service ability; power_dis tribution; safety_pro tection	none	none	none
assumptio n_resoluti on	major	needs review	Secure Managem ent	Can the Secure Managem ent assumption be accepted for architectu re review, or must it be replaced with evidence?	Supply an engineerin g calculation , reviewer note, test plan or datasheet -backed rationale for this envelope.	none	none	none	none
assumptio n_resoluti on	major	needs review	Thermal/P ower Safety	Can the Thermal/P ower Safety assumption be accepted for architectu re review, or must it be replaced with evidence?	Supply an engineerin g calculation , reviewer note, test plan or datasheet -backed rationale for this envelope.	compute_ accelerati on; environme ntal_interf ace; control_co mpute_co mmunicati on; structure_ containme nt; maintenanc e_service ability; power_dis tribution; safety_pro tection	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
calculation_ envelope	major	ready	Rack power density	Does the Rack power density calculation support the architecture envelope, or does the concept need revision?	Confirm inlet temperature, airflow, fan curve, acoustic limit, PSU derating and service clearance.	structure_ containme nt; environme ntal_interf ace; maintenanc e_service ability; power_dis tribution; safety_pro tection	none	edge_ai_p ower_dens ity_w_per_u	none
calculation_ envelope	major	ready	Edge AI accelerator efficiency	Does the Edge AI accelerator efficiency calculation support the architecture envelope, or does the concept need revision?	Confirm workload precision, sustained throughput, accelerator SKU, thermal derating and host overhead.	compute_ accelerati on; environme ntal_interf ace; control_co mpute_co mmunicati on; power_dis tribution; safety_pro tection	none	edge_ai_t ops_per_w att	none
interface_ contract	major	ready	Control Compute Communication -> Compute Acceleration via pcie_fabric	Is the pcie_fabric interface between Control Compute Communication and Compute Acceleration correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	control_co mpute_co mmunicati on; compute_ accelerati on	pcie_fabri c	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_ contract	major	ready	Data Storage -> Control Compute Communication via storage_bus	Is the storage_bus interface between Data Storage and Control Compute Communication correctly owned	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any	data_stora ge; control_co mpute_co mmunicati on	storage_b us	none	Both endpoints expose the required interface through at least one submodule carrier.

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				and physically carried?	deferred checks.				
interface_contract	major	ready	Thermal Airflow Interface -> Compute Acceleration via thermal_path	Is the thermal_path interface between Thermal Airflow Interface and Compute Acceleration correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	environmental_interface; compute_acceleration	thermal_path	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Network IO -> Control Compute Communication via inference_network	Is the inference_network interface between Network IO and Control Compute Communication correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	network_io; control_compute_communication	inference_network	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Power Distribution -> Compute Acceleration via gpu_power_bus	Is the gpu_power_bus interface between Power Distribution and Compute Acceleration correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	power_distribution; compute_acceleration	gpu_power_bus	none	Both endpoints expose the required interface through at least one submodule carrier.

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
interface_contract	major	ready	Security And Safety Protection -> Power Distribution via hardwired_trip	Is the hardwired_trip interface between Security And Safety Protection and Power Distribution correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	safety_protection; power_distribution	hardwired_trip	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Telemetry Instrumentation -> Control Compute Communication via sensor_buses	Is the sensor_buses interface between Telemetry Instrumentation and Control Compute Communication correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	sensing_instrumentation; control_compute_communication	sensor_buses	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Rack Structure Containment -> Thermal Airflow Interface via airflow_path	Is the airflow_path interface between Rack Structure Containment and Thermal Airflow Interface correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	structure_containment; environmental_interface	airflow_path	none	Both endpoints expose the required interface through at least one submodule carrier.
module_allocation	major	ready	Compute Acceleration	Does Compute Acceleration have the right responsibility boundary, submodule split and interface ownership	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	compute_acceleration	pcie_fabric; memory_buses; gpu_power_bus; thermal_path; sensor_buses; low_voltage_rail	edgeai_compute_throughput; edgeai_data_compute_closure; edgeai_thermal_power_safety	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				for this concept?					
module_all ocation	major	ready	Control Compute Communication	Does Control Compute Communication have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	control_compute_communication	pcie_fabric; storage_buses; management_network; inference_network; sensor_buses; thermal_alarm_bus; secure_boot_chain; service_access	edgeai_compute_throughput; edgeai_data_computation_closure; edgeai_thermal_power_safety	none
module_all ocation	major	ready	Data Storage	Does Data Storage have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	data_storage	storage_buses; pcie_fabric; service_access; secure_boot_chain; management_network	none	none
module_all ocation	major	ready	Thermal Airflow Interface	Does Thermal Airflow Interface have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	environmental_interface	airflow_path; thermal_path; fan_control_bus; sensor_buses; service_access	edgeai_compute_throughput; edgeai_data_computation_closure; edgeai_thermal_power_safety; edgeai_power_density	none
module_all ocation	major	ready	Field Maintenance Serviceability	Does Field Maintenance Serviceability have the right responsibility boundary, submodule split and interface ownership	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	maintenance_serviceability	service_access; rack_mount; management_network; airflow_path; secure_boot_chain	edgeai_power_density; edgeai_thermal_power_safety	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				for this concept?					
module_all ocation	major	ready	Network IO	Does Network IO have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	network_io	inference_network; management_network; pcie_fabric; service_access; protective_earth	none	none
module_all ocation	major	ready	Power Distribution	Does Power Distribution have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	power_distribution	ac_input_bus; low_voltage_rail; gpu_power_bus; protective_earth; hardwired_trip; sensor_buses; service_access	edgeai_power_density; edgeai_thermal_power_safety	none
module_all ocation	major	ready	Security And Safety Protection	Does Security And Safety Protection have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	safety_protection	secure_boot_chain; thermal_alarm_bus; hardwired_trip; protective_earth; service_access; management_network; fan_control_bus	edgeai_power_density; edgeai_thermal_power_safety	none
module_all ocation	major	ready	Telemetry Instrumentation	Does Telemetry Instrumentation have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	sensing_instrumentation	sensor_buses; management_network; thermal_alarm_bus; fan_control_bus; service_access	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
module_all ocation	major	ready	Rack Structure Containme nt	Does Rack Structure Containme nt have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	structure_ containme nt	rack_mou nt; airflow_pa th; service_ac cess; protective _earth	edgeai_po wer_densit y; edgeai_th ermal_po wer_safet y	none
submodul e_allocation	major	ready	Compute Accelerati on / Accelerato r module	Do the componen t candidate s and interfaces assigned to Accelerato r module make engineerin g sense for its stated purpose?	Reviewer note confirming the submodule role, componen t allocation and any missing carriers or capacity checks.	compute_ accelerati on	pcie_fabri c; gpu_powe r_bus; thermal_p ath; sensor_bu s	edgeai_co mpute_thr oughput; edgeai_da ta_comput e_closure; edgeai_th ermal_po wer_safet y	none
submodul e_allocation	major	ready	Compute Accelerati on / CPU and memory complex	Do the componen t candidate s and interfaces assigned to CPU and memory complex make engineerin g sense for its stated purpose?	Reviewer note confirming the submodule role, componen t allocation and any missing carriers or capacity checks.	compute_ accelerati on	memory_b us; pcie_fabri c; low_voltag e_rail; sensor_bu s	edgeai_co mpute_thr oughput; edgeai_da ta_comput e_closure; edgeai_th ermal_po wer_safet y	none
submodul e_allocation	major	ready	Compute Accelerati on / Signal and power integrity	Do the componen t candidate s and interfaces assigned to Signal and power integrity make engineerin g sense for its	Reviewer note confirming the submodule role, componen t allocation and any missing carriers or capacity checks.	compute_ accelerati on	pcie_fabri c; gpu_powe r_bus; thermal_p ath	edgeai_co mpute_thr oughput; edgeai_da ta_comput e_closure; edgeai_th ermal_po wer_safet y	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				stated purpose?					
submodule_allocation	major	ready	Control Compute Communication / Host control	Do the component candidates and interfaces assigned to Host control make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	control_compute_communication	pcie_fabric; storage_buses; management_network; sensor_buses	edgeai_compute_throughput; edgeai_data_computation_closure; edgeai_thermal_power_safety	none
submodule_allocation	major	ready	Control Compute Communication / Remote management	Do the component candidates and interfaces assigned to Remote management make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	control_compute_communication	management_network; service_access; secure_boot_chain	edgeai_compute_throughput; edgeai_data_computation_closure; edgeai_thermal_power_safety	none
submodule_allocation	major	ready	Control Compute Communication / Runtime orchestration	Do the component candidates and interfaces assigned to Runtime orchestration make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	control_compute_communication	pcie_fabric; storage_buses; inference_network; secure_boot_chain	edgeai_compute_throughput; edgeai_data_computation_closure; edgeai_thermal_power_safety	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodule_allocation	major	ready	Data Storage / Log and evidence buffer	Do the component candidates and interfaces assigned to Log and evidence buffer make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	data_storage	storage_bu s; managem ent_network; service_access	none	none
submodule_allocation	major	ready	Data Storage / Model cache and recovery	Do the component candidates and interfaces assigned to Model cache and recovery make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	data_storage	storage_bu s; secure_bo ot_chain; managem ent_network	none	none
submodule_allocation	major	ready	Data Storage / NVMe storage	Do the component candidates and interfaces assigned to NVMe storage make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	data_storage	storage_bu s; pcie_fabric; service_access	none	none
submodule_allocation	major	ready	Thermal Airflow Interface / Forced-air cooling	Do the component candidates and interfaces assigned to Forced-air cooling make engineering sense for its	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environmental_interface	airflow_path; fan_control_bus; sensor_bu s	edgeai_compute_throughput; edgeai_data_compute_closure; edgeai_thermal_power_safety; edgeai_power_density	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				stated purpose?					
submodule_allocation	major	ready	Thermal Airflow Interface / Heat transfer path	Do the component candidates and interfaces assigned to Heat transfer path make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environmental_interface	thermal_path; airflow_path; service_access	edgeai_compute_throughput; edgeai_data_computation_closure; edgeai_thermal_power_safety; edgeai_power_density	none
submodule_allocation	major	ready	Thermal Airflow Interface / Inlet and outlet management	Do the component candidates and interfaces assigned to Inlet and outlet management make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environmental_interface	airflow_path; service_access	edgeai_compute_throughput; edgeai_data_computation_closure; edgeai_thermal_power_safety; edgeai_power_density	none
submodule_allocation	major	ready	Field Maintenance Serviceability / Field-replaceable units	Do the component candidates and interfaces assigned to Field-replaceable units make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	maintenance_serviceability	service_access; airflow_path	edgeai_power_density; edgeai_thermal_power_safety	none
submodule_allocation	major	ready	Field Maintenance Serviceability / Rack installation	Do the component candidates and interfaces assigned to Rack installation make engineering sense	Reviewer note confirming the submodule role, component allocation and any missing carriers or	maintenance_serviceability	rack_mount; service_access	edgeai_power_density; edgeai_thermal_power_safety	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				for its stated purpose?	capacity checks.				
submodule_allocation	major	ready	Field Maintenance Serviceability / Service workflow	Do the component candidates and interfaces assigned to Service workflow make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	maintenance_service_ability	service_access; management_network; secure_boot_chain	edgeai_power_density; edgeai_thermal_power_safety	none
submodule_allocation	major	ready	Network IO / High-speed network	Do the component candidates and interfaces assigned to High-speed network make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	network_io	inference_network; pcie_fabric; service_access	none	none
submodule_allocation	major	ready	Network IO / IO isolation and routing	Do the component candidates and interfaces assigned to IO isolation and routing make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	network_io	inference_network; service_access; protective_earth	none	none
submodule_allocation	major	ready	Network IO / Timing and ingest	Do the component candidates and interfaces assigned to Timing and ingest make	Reviewer note confirming the submodule role, component allocation and any	network_io	inference_network; management_network; service_access	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
submodule_allocation	major	ready	Power Distribution / AC input conversion	engineering sense for its stated purpose?	missing carriers or capacity checks.				
				Do the component candidates and interfaces assigned to AC input conversion make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	power_distribution	ac_input_bus; protective_earth; sensor_buses	edgeai_power_density; edgeai_thermal_power_safety	none

Showing first 70 of 81 engineering review questions; export CSV for the full pack.

Engineering Sanity Checks

Check	Status	Value	Expected	Interpretation
Inference Throughput	pass	200 TOPS	10 to 2000 TOPS for a rack edge inference appliance before model and latency derating	200 TOPS is plausible only after model benchmark, precision mode, batching and accelerator thermal derating evidence.
Rack Power Density	pass	700 W/U	100 to 1500 W/U for forced-air edge appliances; higher density needs detailed rack inlet and cooling proof	700 W in 1U implies 700 W/U before rack inlet temperature, airflow and PSU derating evidence.
Data/Compute Closure	pass	present	PCIe fabric, inference network, storage bus and management network present	Accelerator, host control, network ingress and model storage are connected before BoM sourcing.
Thermal/Power Safety	pass	present	thermal path, airflow path, fan control, AC input, protective earth, thermal alarm and hardwired trip present	Rack thermal management and hardware shutdown paths are named before any sourced BoM or cost claim.
Secure Management	pass	present	secure boot chain, management network, service access and TPM present	The architecture includes a measurable secure-management path before fleet deployment claims.

Engineering Calculation Ledger

2 calculations: 2 within deterministic envelope, 0 needing engineering review, 0 outside envelope, 0 blocked by missing inputs. These are arithmetic gates, not proof of real-world performance.

Calculation	Status	Formula	Result	Envelope	Interpretation	Evidence Required
Edge AI accelerator efficiency	within envelope	compute_tops / power_budget_w	0.29 TOPS/W	>=0.1 TOPS/W is a broad plausibility floor before accelerator selection.	0.29 TOPS/W using total appliance power budget.	Confirm workload precision, sustained throughput, accelerator SKU, thermal derating and host overhead.
Rack power density	within envelope	power_budget_w / rack_units	700 W/U	<=1,000 W/U is plausible for dense air-cooled edge equipment before thermal proof.	700 W/U rack density before thermal validation.	Confirm inlet temperature, airflow, fan curve, acoustic limit, PSU derating and service clearance.

Engineering Assumption Ledger

34 assumptions: 3 brief-supported, 8 model-present, 9 needing engineering review, 14 needing source evidence, 0 blocked. Architecture blockers 0; BoM blockers 14.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
Inference throughput	brief_requirement	brief supported	no	no	Inference throughput = 200 TOPS is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Rack height	brief_requirement	brief supported	no	no	Rack height = 1 U is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Power budget	brief_requirement	brief supported	no	no	Power budget = 700 W is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Inference throughput	derived_metric	review required	no	no	Inference throughput uses compute_tops with medium confidence.	Review the formula, input values and operating envelope before treating this metric as a performance claim.
Estimated CAPEX	derived_metric	source required	no	yes	Estimated CAPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
Estimated annual OPEX	derived_metric	source required	no	yes	Estimated annual OPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
Inference Throughput	sanity_envelope	review required	no	no	Inference Throughput currently evaluates as pass against 10 to 2000 TOPS for a rack edge inference appliance before model and latency derating.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
Rack Power Density	sanity_envelope	review required	no	no	Rack Power Density currently evaluates as pass against 100 to 1500 W/U for forced-air edge appliances; higher density needs detailed rack inlet and cooling proof.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
Data/Compute Closure	sanity_envelope	review required	no	no	Data/Compute Closure currently evaluates as pass against PCIe fabric, inference network, storage bus and management network present.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
Thermal/Power Safety	sanity_envelope	review required	no	no	Thermal/Power Safety currently evaluates as pass against thermal path, airflow path, fan control, AC input, protective earth, thermal alarm and hardwired trip present.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
Secure Management	sanity_envelope	review required	no	no	Secure Management currently evaluates as pass against secure boot chain, management network, service access and TPM present.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
control_compute_communication -> compute_acceleration	interface_closure	model present	no	no	pcie_fabric is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
power_distribution -> compute_acceleration	interface_closure	model present	no	no	gpu_power_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
network_io -> control_compute_communication	interface_closure	model present	no	no	inference_network is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
data_storage -> control_compute_communication	interface_closure	model present	no	no	storage_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
environmental_interface -> compute_acceleration	interface_closure	model present	no	no	thermal_path is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
sensing_instrumentation -> control_compute_communication	interface_closure	model present	no	no	sensor_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
safety_protection -> power_distribution	interface_closure	model present	no	no	hardwired_trip is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
structure_containment -> environmental_interface	interface_closure	model present	no	no	airflow_path is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
1U rack chassis	critical_component	source required	no	yes	1U rack chassis is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
AI accelerator module	critical_component	source required	no	yes	AI accelerator module is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
CPU module	critical_component	source required	no	yes	CPU module is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
ECC memory	critical_component	source required	no	yes	ECC memory is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
System motherboard	critical_component	source required	no	yes	System motherboard is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
High-speed NIC	critical_component	source required	no	yes	High-speed NIC is a critical architecture-derived	Admit supplier, manufacturer, MPN where applicable, unit

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					component candidate with quantity 1 candidate.	cost, lead time and evidence reference through sourcing intake.
NVMe SSD	critical_component	source required	no	yes	NVMe SSD is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
BMC management controller	critical_component	source required	no	yes	BMC management controller is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Redundant power supply	critical_component	source required	no	yes	Redundant power supply is a critical architecture-derived component candidate with quantity 2 each.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Fan wall assembly	critical_component	source required	no	yes	Fan wall assembly is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Heatsink cold plate	critical_component	source required	no	yes	Heatsink cold plate is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
TPM security module	critical_component	source required	no	yes	TPM security module is a critical architecture-derived component candidate with	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					quantity 1 candidate.	through sourcing intake.
IEC-62368-1	compliance_evidence	review required	no	no	Audio/video, ICT and communication equipment safety is relevant to this product class in UK/EU/global.	Electrical safety, enclosure, temperature and abnormal-operation evidence.
CE-EMC	compliance_evidence	review required	no	no	Electromagnetic compatibility technical file is relevant to this product class in UK/EU.	EMC design review, pre-compliance or test evidence for rack appliance.
SECURE-BOOT-REVIEW	compliance_evidence	review required	no	no	Secure boot and fleet-management review is relevant to this product class in Project security gate.	TPM, measured boot, firmware signing, recovery and management-access evidence.

Engineering Verification Plan

39 activities: 25 ready for review, 6 open, 8 blocked.

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review Rack Structure Containment architecture allocation	ready	design_review	Rack Structure Containment	rack_units	rack_mount ; airflow_path ; service_access; protective_earth	1u_rack_chassis; slide_rail_kit; internal_divider_panel; front_bezel_assembly; accelerator_retention_bracket; pcie_riser_support; card_stiffener_rail; gpu_airflow_baffle; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Rack Structure Containment requirement calculations	ready	calculation	Rack Structure Containment	rack_units	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Collect source-backed evidence for Rack Structure Containment critical components	blocked	source_evidence	Rack Structure Containment	rack_units	none	1u_rack_chassis	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved	1u_rack_chassis

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	
Review Compute Acceleration architecture allocation	ready	design_review	Compute Acceleration	compute_tops	pcie_fabric; memory_buses; gpu_power_bus; thermal_patching; sensor_bus; low_voltage_rail	ai_accelerator_module; accelerator_carrier_board; pcie_edge_connector; accelerator_retention_latch; cpu_module; ecc_memory; system_motherboard; boot_flash_device; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Compute Acceleration requirement calculations	ready	calculation	Compute Acceleration	compute_tops	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Collect source-backed evidence for Compute Acceleration critical components	blocked	source_evidence	Compute Acceleration	compute_tops	none	ai_accelerator_module; cpu_module; ecc_memory; system_motherboard	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	ai_accelerator_module cpu_module ecc_memory system_motherboard
Review Network IO architecture allocation	ready	design_review	Network IO	none	inference_network; management_network; pcie_fabric; service_access; protective_earth	high_speed_nic; optical_transceiver_cage; nic_heatsink; network_link_led_board; ptp_timing_module; gpio_trigger_input; data_ingest_connector; network_by_pass_relay; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Collect source-backed evidence for Network IO critical components	blocked	source_evidence	Network IO	none	none	high_speed_nic	Supplier name, manufacturer or equivalent, MPN where applicable,	high_speed_nic

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	
Review Data Storage architecture allocation	ready	design_review	Data Storage	none	storage_bus; pcie_fabric; service_access; secure_boot_chain; management_network	nvme_ssd; m_2_carrier_tray; drive_thermal_pad; drive_retention_screw_set; model_cache_partition; storage_health_monitor; secure_eraser_controller; firmware_recovery_image; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Collect source-backed evidence for Data Storage critical components	blocked	source_evidence	Data Storage	none	none	nvme_ssd	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved	nvme_ssd

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	
Review Control Compute Communication architecture allocation	ready	design_review	Control Compute Communication	compute_tops	pcie_fabric; storage_bus; management_network; inference_network; sensor_bus; thermal_alarm_bus; secure_boot_chain; service_access	bmc_management_controller; management_ethernet_phy; bios_firmware_image; platform_configuration_eeprom; inference_runtime_image; accelerator_driver_bundle; watchdog_service; configuration_manifest; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Control Compute Communication requirement calculations	ready	calculation	Control Compute Communication	compute_tops	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Collect source-backed evidence for Control Compute Communication critical components	blocked	source_evidence	Control Compute Communication	compute_tops	none	bmc_management_controller	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	bmc_management_controller
Review Power Distribution architecture allocation	ready	design_review	Power Distribution	power_budget_w	ac_input_buses; low_voltage_rail; gpu_power_bus; protective_earth; hardwired_trip; sensor_bus; service_access	redundant_power_supply; ac_inlet_filter; input_fuse_holder; power_good_signal_harness; 12_v_busbar; accelerator_power_harness; hot_swap_controller; fan_power_distribution_loom; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Power Distribution requirement calculations	ready	calculation	Power Distribution	power_budget_w	none	none	Each linked requirement has a calculation, sanity check or clearly	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	
Collect source-backed evidence for Power Distribution critical components	blocked	source_evidence	Power Distribution	power_budget_w	none	redundant_power_supply	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	redundant_power_supply
Review Thermal Airflow Interface architecture allocation	ready	design_review	Thermal Airflow Interface	compute_tops; rack_units; power_budget_w	airflow_path; thermal_path; fan_control_bus; sensor_bus; service_access	fan_wall_assembly; pwm_fan_controller; airflow_straightener; replaceable_dust_filter; heatsink_cold_plate; thermal_interface_material_set; heatpipe_assembly; rear_exhaust_duct; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							No blocker or major architecture issue is attached to the module.	
Verify Thermal Airflow Interface requirement calculations	ready	calculation	Thermal Airflow Interface	compute_tops; rack_units; power_budget_w	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Collect source-backed evidence for Thermal Airflow Interface critical components	blocked	source_evidence	Thermal Airflow Interface	compute_tops; rack_units; power_budget_w	none	fan_wall_assembly; heatsink_cold_plate	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	fan_wall_assembly heatsink_cold_plate

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review Telemetry Instrumentation architecture allocation	ready	design_review	Telemetry Instrumentation	none	sensor_bus; management_network; thermal_alarm_bus; fan_control_bus; service_access	inlet_temperature_sensor; accelerator_hotspot_sensor; exhaust_temperature_sensor; board_humidity_sensor; psu_pmbus_monitor; rail_current_monitor; fan_tachometer_hub; chassis_intrusion_switch; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Review Security And Safety Protection architecture allocation	ready	design_review	Security And Safety Protection	power_budget_w	secure_boot_chain; thermal_alarm_bus; hardwired_trip; protective_earth; service_access; management_network; fan_control_bus	tpm_security_module; secure_boot_policy; chassis_tamper_switch; recovery_jumper_cover; thermal_trip_relay; overtemperature_latch; fan_failure_interlock; hardwired_shutdown_line; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Security And Safety Protection requirement calculations	ready	calculation	Security And Safety Protection	power_budget_w	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	
Collect source-backed evidence for Security And Safety Protection critical components	blocked	source_evidence	Security And Safety Protection	power_budget_w	none	tpm_security_module	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	tpm_security_module
Review Field Maintenance Serviceability architecture allocation	ready	design_review	Field Maintenance Serviceability	rack_units	service_access; rack_mount; management_network; airflow_path; secure_boot_chain	hot_swap_fan_carrier; drive_service_handle; psu_latch_mechanism; accelerator_extraction_handle; rack_rail_kit; rear_support_bracket; cable_retention_comb; airflow_blinking_panel; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							issue is attached to the module.	
Verify Field Maintenance Serviceability requirement calculations	ready	calculation	Field Maintenance Serviceability	rack_units	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Verify pcie_fabric interface contract	ready	interface_review	Control Compute Communication -> Compute Acceleration	none	pcie_fabric	bmc_management_controller; management_ethernet_phy; bios_firmware_image; platform_configuration_eeprom; inference_runtime_image; accelerator_driver_bundle; watchdog_service; configuration_manifest; plus 12 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify gpu_power_bus interface contract	ready	interface_review	Power Distribution -> Compute Acceleration	none	gpu_power_bus	12_v_busbar; accelerator_power_harness; hot_swap_controller; fan_power_distribution_loom; ai_accelerator_module; accelerator_carrier_board;	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						pcie_edge_connector; accelerator_retention_latch; plus 4 more	or data capacity is validated by the next detailed design pass.	
Verify inference_network interface contract	ready	interface_review	Network IO -> Control Compute Communication	none	inference_network	high_speed_nic; optical_transceiver_cage; nic_heatsink; network_link_led_board; ptp_timing_module; gpio_trigger_input; data_ingest_connector; network_by_pass_relay; plus 8 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify storage_bus interface contract	ready	interface_review	Data Storage -> Control Compute Communication	none	storage_bus	nvme_ssd; m_2_carrier_tray; drive_thermal_pad; drive_retention_screw_set; model_cache_partition; storage_health_monitor; secure_erase_controller; firmware_recovery_image; plus 12 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify thermal_path interface contract	ready	interface_review	Thermal Airflow Interface -> Compute Acceleration	none	thermal_path	heatsink_cold_plate; thermal_interface_material_set; heatpipe_assembly; rear_exhaust_duct; ai_accelerator_module;	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						accelerator_carrier_board; pcie_edge_connector; accelerator_retention_latch; plus 4 more	Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	
Verify sensor_bus interface contract	ready	interface_review	Telemetry Instrumentation -> Control Compute Communication	none	sensor_bus	inlet_temperature_sensor; accelerator_hotspot_sensor; exhaust_temperature_sensor; board_humidity_sensor; psu_pmbus_monitor; rail_current_monitor; fan_tachometer_hub; chassis_intrusion_switch; plus 8 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify hardwired_trip interface contract	ready	interface_review	Security And Safety Protection -> Power Distribution	none	hardwired_trip	thermal_trip_relay; overtemperature_latch; fan_failure_interlock; hardwired_shutdown_line; protective_earth_stud; insulation_barrier_sheet; finger_safe_psu_cover; warning_label_set; plus 4 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Verify airflow_path interface contract	ready	interface_review	Rack Structure Containment -> Thermal Airflow Interface	none	airflow_path	1u_rack_chassis; slide_rail_kit; internal_divider_panel; front_bezel_assembly; accelerator_retention_bracket; pcie_riser_support; card_stiffener_rail; gpu_airflow_baffle; plus 12 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Plan evidence for IEC-62368-1	open	compliance_review	System Compliance	none	none	none	Electrical safety, enclosure, temperature and abnormal-operation evidence.	Compliance evidence not yet collected in scratch prototype.
Plan evidence for CE-EMC	open	compliance_review	System Compliance	none	none	none	EMC design review, pre-compliance or test evidence for rack appliance.	Compliance evidence not yet collected in scratch prototype.
Plan evidence for SECURE-BOOT-REVIEW	open	compliance_review	System Compliance	none	none	none	TPM, measured boot, firmware signing, recovery and management-access evidence.	Compliance evidence not yet collected in scratch prototype.
Review risk control for Accelerator thermal throttling or shutdown under rack inlet conditions	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted	Risk evidence not yet reviewed in scratch prototype.

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							into a design action. Any residual high-risk item has an owner before publication.	
Review risk control for Insecure firmware or management interface compromise	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	Risk evidence not yet reviewed in scratch prototype.
Review risk control for Inference outage from PSU, fan or SSD failure	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	Risk evidence not yet reviewed in scratch prototype.

Verification Evidence Ledger

0 accepted, 0 rejected, 0 deferred, 31 pending, 0 blocked without evidence. Review coverage 0; acceptance ratio 0. Source-evidence activities remain in sourcing intake.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Review Rack Structure Containment architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Rack Structure Containment requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Rack Structure Containment critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Compute Acceleration architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Compute Acceleration requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Compute Acceleration critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Network IO architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Network IO critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Data Storage architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Data Storage critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Control Compute Communication architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Control Compute Communication	pending	calculation			Collect reviewer evidence through

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
requirement calculations					the verification intake template.
Collect source-backed evidence for Control Compute Communication critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Power Distribution architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Power Distribution requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Power Distribution critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Thermal Airflow Interface architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Thermal Airflow Interface requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Thermal Airflow Interface critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Telemetry Instrumentation architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Review Security And Safety Protection architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Security And Safety Protection requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Security And Safety Protection critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Field Maintenance	pending	design_review			Collect reviewer evidence through

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Serviceability architecture allocation					the verification intake template.
Verify Field Maintenance Serviceability requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Verify pcie_fabric interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify gpu_power_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify inference_network interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify storage_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify thermal_path interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify sensor_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify hardwired_trip interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify airflow_path interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Plan evidence for IEC-62368-1	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Plan evidence for CE-EMC	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Plan evidence for SECURE-BOOT- REVIEW	pending	compliance_review			Collect reviewer evidence through the verification intake template.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Review risk control for Accelerator thermal throttling or shutdown under rack inlet conditions	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for Insecure firmware or management interface compromise	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for Inference outage from PSU, fan or SSD failure	pending	compliance_review			Collect reviewer evidence through the verification intake template.

Rack Structure Containment

Provides a 1U rack envelope with controlled airflow, card retention and service access.

- rack_mount
- airflow_path
- service_access
- protective_earth

Rack chassis

Carries appliance boards, power supplies and front-to-back airflow features.

Components: 1U rack chassis, slide rail kit, internal divider panel, front bezel assembly

Accelerator mounting

Restrains high-power cards and maintains airflow clearance.

Components: accelerator retention bracket, PCIe riser support, card stiffener rail, GPU airflow baffle

Service labelling

Marks rack identity, airflow direction and field-replaceable units.

Components: asset label plate, airflow direction label, rack ear set, cable management arm

Compute Acceleration

Provides roughly 200 TOPS of accelerator throughput before model and thermal derating evidence.

- pcie_fabric
- memory_bus
- gpu_power_bus
- thermal_path
- sensor_bus

Accelerator module

Runs neural-network inference workloads at the edge.

Components: AI accelerator module, accelerator carrier board, PCIe edge connector, accelerator retention latch

CPU and memory complex

Hosts preprocessing, runtime orchestration and memory-resident model state.

Components: CPU module, ECC memory, System motherboard, boot flash device

Signal and power integrity

Stabilises accelerator clocks, high-speed lanes and transient power demand.

Components: high-current VRM module, PCIe retimer set, clock distribution device, power integrity capacitor bank

Network IO

Moves inference requests, responses and timing data through the appliance boundary.



High-speed network

Terminates production inference traffic.

Components: High-speed NIC, optical transceiver cage, NIC heatsink, network link LED board

Timing and ingest

Synchronises inference streams and accepts trigger inputs where needed.

Components: PTP timing module, GPIO trigger input, data ingest connector, network bypass relay

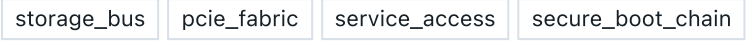
IO isolation and routing

Controls EMC, strain relief and service routing at high-speed ports.

Components: EMI gasket strip, shielded IO bracket, cable strain relief comb, service loop tie set

Data Storage

Stores models, runtime images, logs and local inference buffers.



NVMe storage

Provides local high-throughput storage for models and telemetry.

Components: NVMe SSD, M.2 carrier tray, drive thermal pad, drive retention screw set

Model cache and recovery

Keeps model versions, rollback images and erase functions controlled.

Components: model cache partition, storage health monitor, secure erase controller, firmware recovery image

Log and evidence buffer

Captures operational evidence for debugging and compliance review.

Components: event log partition, write endurance counter, crash dump region, log export port

Control Compute Communication

Runs host firmware, inference orchestration, management control and secure remote operation.



Host control

Coordinates CPU, accelerator, storage and management functions.

Components: BMC management controller, management Ethernet PHY, BIOS firmware image, platform configuration EEPROM

Runtime orchestration

Loads inference runtime, accelerator drivers and service policy.

Components: inference runtime image, accelerator driver bundle, watchdog service, configuration manifest

Remote management

Supports controlled diagnostics and firmware maintenance.

Components: out-of-band management port, serial console header, debug access lockout, firmware signing key slot

Power Distribution

Converts AC input into protected appliance rails for a 700 W power-budget target.

ac_input_bus	low_voltage_rail	gpu_power_bus	protective_earth	hardwired_trip	sensor_bus
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AC input conversion

Accepts rack AC input and provides redundant conversion.

Components: Redundant power supply, AC inlet filter, input fuse holder, power-good signal harness

DC rail distribution

Distributes 12 V and accelerator rails to boards and fans.

Components: 12 V busbar, accelerator power harness, hot-swap controller, fan power distribution loom

Hold-up and protection

Survives short dips and trips unsafe rail faults.

Components: hold-up capacitor module, brownout detector, current shunt module, eFuse channel set

Thermal Airflow Interface

Controls front-to-back airflow and heat transfer from accelerator, CPU, NIC and storage.

airflow_path	thermal_path	fan_control_bus	sensor_bus	service_access
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Forced-air cooling

Maintains rack airflow through compute and power zones.

Components: Fan wall assembly, PWM fan controller, airflow straightener, replaceable dust filter

Heat transfer path

Couples accelerator and CPU losses into chassis airflow.

Components: Heatsink cold plate, thermal interface material set, heatpipe assembly, rear exhaust duct

Inlet and outlet management

Protects rack pressure and prevents hot-air recirculation.

Components: front inlet grille, rear exhaust grille, air seal foam kit, rack pressure map label

Telemetry Instrumentation

Measures temperature, fan, power, intrusion and service state for health management.

sensor_bus	management_network	thermal_alarm_bus	fan_control_bus
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Thermal sensors

Measures inlet, accelerator, exhaust and board climate state.

Components: inlet temperature sensor, accelerator hotspot sensor, exhaust temperature sensor, board humidity sensor

Power and fan telemetry

Feeds PSU, rail and airflow state into BMC management.

Components: PSU PMBus monitor, rail current monitor, fan tachometer hub, chassis intrusion switch

Front panel status

Exposes health and service state to rack technicians.

Components: front status LED board, UID button, alarm buzzer, LCD status panel

Security And Safety Protection

Protects boot integrity, operator safety and thermal shutdown paths.

secure_boot_chain	thermal_alarm_bus	hardwired_trip	protective_earth	service_access
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Secure boot

Anchors platform identity and firmware trust.

Components: TPM security module, secure boot policy, chassis tamper switch, recovery jumper cover

Thermal shutdown

Trips unsafe thermal or fan-failure states.

Components: thermal trip relay, overtemperature latch, fan failure interlock, hardwired shutdown line

Electrical safety

Maintains operator protection and safe service boundaries.

Components: protective earth stud, insulation barrier sheet, finger-safe PSU cover, warning label set

Field Maintenance Serviceability

Supports rack installation, hot-swap FRUs, diagnostics and firmware recovery.

service_access	rack_mount	management_network	airflow_path
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Field-replaceable units

Makes high-wear parts replaceable without removing the whole appliance.

Components: hot-swap fan carrier, drive service handle, PSU latch mechanism, accelerator extraction handle

Rack installation

Maintains mechanical support and cabling in edge racks.

Components: rack rail kit, rear support bracket, cable retention comb, airflow blanking panel

Service workflow

Captures field diagnostics and controlled recovery evidence.

Components: service checklist card, firmware version label, spare screw kit, diagnostics USB key

Bill of Materials

Pricing status: architecture-derived candidate BoM only. Costs, suppliers, manufacturers and part numbers have not been independently sourced.

Total CAPEX BoM: £0

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
1U rack chassis	1	candidate	unpriced	unpriced				
slide rail kit	1	candidate	unpriced	unpriced				
internal divider panel	1	candidate	unpriced	unpriced				
front bezel assembly	1	candidate	unpriced	unpriced				
accelerator retention bracket	1	candidate	unpriced	unpriced				
PCIe riser support	1	candidate	unpriced	unpriced				
card stiffener rail	1	candidate	unpriced	unpriced				
GPU airflow baffle	1	candidate	unpriced	unpriced				
asset label plate	1	candidate	unpriced	unpriced				
airflow direction label	1	candidate	unpriced	unpriced				
rack ear set	1	candidate	unpriced	unpriced				
cable management arm	1	candidate	unpriced	unpriced				
AI accelerator module	1	candidate	unpriced	unpriced				
accelerator carrier board	1	candidate	unpriced	unpriced				
PCIe edge connector	1	candidate	unpriced	unpriced				
accelerator retention latch	1	candidate	unpriced	unpriced				
CPU module	1	candidate	unpriced	unpriced				
ECC memory	1	candidate	unpriced	unpriced				
System motherboard	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
boot flash device	1	candidate	unpriced	unpriced				
high-current VRM module	1	candidate	unpriced	unpriced				
PCIe retimer set	1	candidate	unpriced	unpriced				
clock distribution device	1	candidate	unpriced	unpriced				
power integrity capacitor bank	1	candidate	unpriced	unpriced				
High-speed NIC	1	candidate	unpriced	unpriced				
optical transceiver cage	1	candidate	unpriced	unpriced				
NIC heatsink	1	candidate	unpriced	unpriced				
network link LED board	1	candidate	unpriced	unpriced				
PTP timing module	1	candidate	unpriced	unpriced				
GPIO trigger input	1	candidate	unpriced	unpriced				
data ingest connector	1	candidate	unpriced	unpriced				
network bypass relay	1	candidate	unpriced	unpriced				
EMI gasket strip	1	candidate	unpriced	unpriced				
shielded IO bracket	1	candidate	unpriced	unpriced				
cable strain relief comb	1	candidate	unpriced	unpriced				
service loop tie set	1	candidate	unpriced	unpriced				
NVMe SSD	1	candidate	unpriced	unpriced				
M.2 carrier tray	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
drive thermal pad	1	candidate	unpriced	unpriced				
drive retention screw set	1	candidate	unpriced	unpriced				
model cache partition	1	candidate	unpriced	unpriced				
storage health monitor	1	candidate	unpriced	unpriced				
secure erase controller	1	candidate	unpriced	unpriced				
firmware recovery image	1	candidate	unpriced	unpriced				
event log partition	1	candidate	unpriced	unpriced				
write endurance counter	1	candidate	unpriced	unpriced				
crash dump region	1	candidate	unpriced	unpriced				
log export port	1	candidate	unpriced	unpriced				
BMC management controller	1	candidate	unpriced	unpriced				
management Ethernet PHY	1	candidate	unpriced	unpriced				
BIOS firmware image	1	candidate	unpriced	unpriced				
platform configuration EEPROM	1	candidate	unpriced	unpriced				
inference runtime image	1	candidate	unpriced	unpriced				
accelerator driver bundle	1	candidate	unpriced	unpriced				
watchdog service	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
configuration manifest	1	candidate	unpriced	unpriced				
out-of-band management port	1	candidate	unpriced	unpriced				
serial console header	1	candidate	unpriced	unpriced				
debug access lockout	1	candidate	unpriced	unpriced				
firmware signing key slot	1	candidate	unpriced	unpriced				
Redundant power supply	2	each	unpriced	unpriced				
AC inlet filter	1	candidate	unpriced	unpriced				
input fuse holder	1	candidate	unpriced	unpriced				
power-good signal harness	1	candidate	unpriced	unpriced				
12 V busbar	1	candidate	unpriced	unpriced				
accelerator power harness	1	candidate	unpriced	unpriced				
hot-swap controller	1	candidate	unpriced	unpriced				
fan power distribution loom	1	candidate	unpriced	unpriced				
hold-up capacitor module	1	candidate	unpriced	unpriced				
brownout detector	1	candidate	unpriced	unpriced				
current shunt module	1	candidate	unpriced	unpriced				
eFuse channel set	1	candidate	unpriced	unpriced				
Fan wall assembly	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
PWM fan controller	1	candidate	unpriced	unpriced				
airflow straightener	1	candidate	unpriced	unpriced				
replaceable dust filter	1	candidate	unpriced	unpriced				
Heatsink cold plate	1	candidate	unpriced	unpriced				
thermal interface material set	1	candidate	unpriced	unpriced				
heatpipe assembly	1	candidate	unpriced	unpriced				
rear exhaust duct	1	candidate	unpriced	unpriced				
front inlet grille	1	candidate	unpriced	unpriced				
rear exhaust grille	1	candidate	unpriced	unpriced				
air seal foam kit	1	candidate	unpriced	unpriced				
rack pressure map label	1	candidate	unpriced	unpriced				
inlet temperature sensor	1	candidate	unpriced	unpriced				
accelerator hotspot sensor	1	candidate	unpriced	unpriced				
exhaust temperature sensor	1	candidate	unpriced	unpriced				
board humidity sensor	1	candidate	unpriced	unpriced				
PSU PMBus monitor	1	candidate	unpriced	unpriced				
rail current monitor	1	candidate	unpriced	unpriced				
fan tachometer hub	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
chassis intrusion switch	1	candidate	unpriced	unpriced				
front status LED board	1	candidate	unpriced	unpriced				
UID button	1	candidate	unpriced	unpriced				
alarm buzzer	1	candidate	unpriced	unpriced				
LCD status panel	1	candidate	unpriced	unpriced				
TPM security module	1	candidate	unpriced	unpriced				
secure boot policy	1	candidate	unpriced	unpriced				
chassis tamper switch	1	candidate	unpriced	unpriced				
recovery jumper cover	1	candidate	unpriced	unpriced				
thermal trip relay	1	candidate	unpriced	unpriced				
overtemperature latch	1	candidate	unpriced	unpriced				
fan failure interlock	1	candidate	unpriced	unpriced				
hardwired shutdown line	1	candidate	unpriced	unpriced				
protective earth stud	1	candidate	unpriced	unpriced				
insulation barrier sheet	1	candidate	unpriced	unpriced				
finger-safe PSU cover	1	candidate	unpriced	unpriced				
warning label set	1	candidate	unpriced	unpriced				
hot-swap fan carrier	1	candidate	unpriced	unpriced				
drive service handle	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
PSU latch mechanism	1	candidate	unpriced	unpriced				
accelerator extraction handle	1	candidate	unpriced	unpriced				
rack rail kit	1	candidate	unpriced	unpriced				
rear support bracket	1	candidate	unpriced	unpriced				
cable retention comb	1	candidate	unpriced	unpriced				
airflow blanking panel	1	candidate	unpriced	unpriced				
service checklist card	1	candidate	unpriced	unpriced				
firmware version label	1	candidate	unpriced	unpriced				
spare screw kit	1	candidate	unpriced	unpriced				
diagnostics USB key	1	candidate	unpriced	unpriced				

Cost And Sourcing

Sourcing Admission

Status	Candidate Lines	Admitted Prices	Unpriced Lines	Unpriced Critical Lines
not_started	120	0	120	12

Critical Sourcing Worklist

Component	Qty	Reason
1U rack chassis	1 candidate	Required class-critical line must be sourced before BoM can pass.
AI accelerator module	1 candidate	Required class-critical line must be sourced before BoM can pass.
CPU module	1 candidate	Required class-critical line must be sourced before BoM can pass.
ECC memory	1 candidate	Required class-critical line must be sourced before BoM can pass.
System motherboard	1 candidate	Required class-critical line must be sourced before BoM can pass.
High-speed NIC	1 candidate	Required class-critical line must be sourced before BoM can pass.
NVMe SSD	1 candidate	Required class-critical line must be sourced before BoM can pass.
BMC management controller	1 candidate	Required class-critical line must be sourced before BoM can pass.
Redundant power supply	2 each	Required class-critical line must be sourced before BoM can pass.
Fan wall assembly	1 candidate	Required class-critical line must be sourced before BoM can pass.
Heatsink cold plate	1 candidate	Required class-critical line must be sourced before BoM can pass.
TPM security module	1 candidate	Required class-critical line must be sourced before BoM can pass.

No rejected source records.

BoM Sourcing Ledger

0/120 priced lines admitted. Critical coverage 0/12. Rejected evidence records: 0. Cost/manufacturer/MPN fields stay blank until evidence is admitted.

Component identity check: no duplicate componentWorldId groups detected.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
1U rack chassis	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture r/MPN/unit -cost evidence before BoM can pass.
slide rail kit	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
internal divider panel	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
front bezel assembly	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
accelerator retention bracket	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
PCIe riser support	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									planning starts.
card stiffener rail	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
GPU airflow baffle	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
asset label plate	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
airflow direction label	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
rack ear set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
cable management arm	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									procurement planning starts.
AI accelerator module	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
accelerator carrier board	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
PCIe edge connector	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
accelerator retention latch	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
CPU module	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
ECC memory	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									r/MPN/unit -cost evidence before BoM can pass.
System motherboard	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
boot flash device	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
high-current VRM module	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
PCIe retimer set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
clock distribution device	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
power integrity capacitor bank	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
High-speed NIC	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
optical transceiver cage	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
NIC heatsink	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
network link LED board	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
PTP timing module	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									planning starts.
GPIO trigger input	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
data ingest connector	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
network bypass relay	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
EMI gasket strip	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
shielded IO bracket	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
cable strain relief comb	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									procurement planning starts.
service loop tie set	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
NVMe SSD	criticalunpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture r/MPN/unit -cost evidence before BoM can pass.
M.2 carrier tray	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
drive thermal pad	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
drive retention screw set	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.

Showing first 40 of 120 ledger rows; export CSV for the full line-by-line ledger.

0 source-backed claims, 48 missing-source claims, 0 provenance violations. Critical missing-source claims: 48. Supplier, manufacturer, MPN and cost fields must appear here with source refs before procurement use.

Line	Claim Field	Status	Value	Source Ref	Next Action
1U rack chassis	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
1U rack chassis	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
1U rack chassis	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
1U rack chassis	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
AI accelerator module	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
AI accelerator module	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
AI accelerator module	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
AI accelerator module	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
CPU module	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
CPU module	manufacturer	missing source			Admit source-backed

Line	Claim Field	Status	Value	Source Ref	Next Action
					supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
CPU module	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
CPU module	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
ECC memory	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
ECC memory	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
ECC memory	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
ECC memory	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
System motherboard	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
System motherboard	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
System motherboard	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost

Line	Claim Field	Status	Value	Source Ref	Next Action
					evidence before BoM can pass.
System motherboard	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
High-speed NIC	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
High-speed NIC	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
High-speed NIC	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
High-speed NIC	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
NVMe SSD	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
NVMe SSD	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
NVMe SSD	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
NVMe SSD	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Line	Claim Field	Status	Value	Source Ref	Next Action
BMC management controller	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
BMC management controller	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
BMC management controller	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
BMC management controller	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Redundant power supply	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Redundant power supply	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Redundant power supply	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Redundant power supply	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Fan wall assembly	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Fan wall assembly	manufacturer	missing source			Admit source-backed

Line	Claim Field	Status	Value	Source Ref	Next Action
					supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Fan wall assembly	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Fan wall assembly	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Heatsink cold plate	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Heatsink cold plate	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Heatsink cold plate	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Heatsink cold plate	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
TPM security module	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
TPM security module	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
TPM security module	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost

Line	Claim Field	Status	Value	Source Ref	Next Action
					evidence before BoM can pass.
TPM security module	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Source Reference Quality Gate

No sourcing evidence references have been admitted. Manufacturer, MPN and cost fields must remain blank until source-backed evidence is admitted through sourcing intake.

BoM Evidence Trace Matrix

0/120 BoM row(s) are production-eligible; 0 protocol-only, 0 source-reference blocked, 12 critical unsourced. Priced-review rows 0; procurement use no; next row bom-trace:structure_containment_rack_chassis_1u_rack_chassis.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
1U rack chassis	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
slide rail kit	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
internal divider panel	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
front bezel assembly	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
accelerator retention bracket	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
PCIe riser support	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									or procurement planning starts.
card stiffener rail	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
GPU airflow baffle	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
asset label plate	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
airflow direction label	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
rack ear set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									planning starts.
cable management arm	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
AI accelerator module	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
accelerator carrier board	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
PCIe edge connector	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
accelerator retention latch	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
CPU module	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									evidence for this critical line before BoM can pass.
ECC memory	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
System motherboard	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
boot flash device	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
high-current VRM module	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
PCIe retimer set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
clock distribution device	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
power integrity capacitor bank	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
High-speed NIC	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
optical transceiver cage	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
NIC heatsink	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
network link LED board	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									or procurement planning starts.
PTP timing module	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
GPIO trigger input	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
data ingest connector	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
network bypass relay	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
EMI gasket strip	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									planning starts.
shielded IO bracket	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
cable strain relief comb	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
service loop tie set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
NVMe SSD	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
M.2 carrier tray	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
drive thermal pad	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									critical sourcing is complete or procurement planning starts.
drive retention screw set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
model cache partition	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
storage health monitor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
secure erase controller	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
firmware recovery image	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									procurement planning starts.
event log partition	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
write endurance counter	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
crash dump region	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
log export port	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
BMC management controller	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
management Ethernet PHY	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Showing first 50 of 120 BoM trace rows; export CSV for the full trace.

BoM Evidence Closure Plan

120 closure row(s): 12 ready, 0 blocked, 108 deferred. Actions: 12 collect source, 0 repair, 0 replace protocol, 108 candidate deferrals. Procurement blockers 12; priced-review blockers 12; next row bom-close:compute_acceleration_accelerator_module_ai_accelerator_module.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
1	AI accelerator module	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
2	CPU module	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
3	ECC memory	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
4	System motherboard	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
5	BMC management controller	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
6	NVMe SSD	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
7	Fan wall assembly	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
8	Heatsink cold plate	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
9	High-speed NIC	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
10	Redundant power supply	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
11	TPM security module	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
12	1U rack chassis	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
13	accelerator carrier board	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
14	accelerator retention latch	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
15	PCIe edge connector	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
16	boot flash device	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
17	clock distribution device	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
18	high-current VRM module	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
19	PCIe retimer set	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
20	power integrity capacitor bank	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
21	BIOS firmware image	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
22	management Ethernet PHY	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
23	platform configuration on EEPROM	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
24	debug access lockout	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
25	firmware signing key slot	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
26	out-of-band management port	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
27	serial console header	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
28	accelerator driver bundle	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
29	configuration manifest	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
30	inference runtime image	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
31	watchdog service	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
32	crash dump region	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
33	event log partition	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
34	log export port	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
35	write endurance counter	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
36	firmware recovery image	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
37	model cache partition	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
38	secure erase controller	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
39	storage health monitor	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
40	drive retention screw set	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
41	drive thermal pad	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
42	M.2 carrier tray	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
43	airflow straightener	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
44	PWM fan controller	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
45	replaceable dust filter	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
46	heatpipe assembly	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
47	rear exhaust duct	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
48	thermal interface material set	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
49	air seal foam kit	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
50	front inlet grille	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Showing first 50 of 120 BoM closure rows; export CSV for the full queue.

Sourcing Batch Plan

2 batch(es): 1 active, 0 waiting, 1 deferred. Active rows 12; deferred rows 108. Critical source rows 12; repair rows 0; protocol replacements 0. Next batch sourcing-batch:2:critical_source_collection; next item critical_source_collection:1:ai_accelerator_module.

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
2	Collect Critical Source Evidence	active	blocker	AI accelerator module	AI accelerator module; AI accelerator module manufacturer part number; AI accelerator module catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	CPU module	CPU module; CPU module manufacturer part number; CPU module catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	ECC memory	ECC memory; ECC memory manufacturer part number; ECC memory catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	System motherboard	System motherboard; System motherboard manufacturer part number; System motherboard catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	BMC management controller	BMC management controller; BMC management controller manufacturer part number; BMC management controller catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
2	Collect Critical Source Evidence	active	blocker	NVMe SSD	NVMe SSD; NVMe SSD manufacturer part number; NVMe SSD catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Fan wall assembly	Fan wall assembly; Fan wall assembly manufacturer part number; Fan wall assembly catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Heatsink cold plate	Heatsink cold plate; Heatsink cold plate manufacturer part number; Heatsink cold plate catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	High-speed NIC	High-speed NIC; High-speed NIC manufacturer part number; High-speed NIC catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Redundant power supply	Redundant power supply; Redundant power supply manufacturer part number; Redundant power supply catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
2	Collect Critical Source Evidence	active	blocker	TPM security module	TPM security module; TPM security module manufacturer part number; TPM security module catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	1U rack chassis	1U rack chassis; 1U rack chassis manufacturer part number; 1U rack chassis catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
6	Defer Candidate-Only Sourcing	deferred	minor	accelerator carrier board	accelerator carrier board; accelerator carrier board manufacturer part number; accelerator carrier board catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	accelerator retention latch	accelerator retention latch; accelerator retention latch manufacturer part number; accelerator retention latch catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	PCIe edge connector	PCIe edge connector; PCIe edge connector manufacturer part number; PCIe edge	no immediate evidence required unless this candidate is promoted to	no	candidate remains out of procurement totals

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
					connector catalogue price GBP	procurement scope		
6	Defer Candidate-Only Sourcing	deferred	minor	boot flash device	boot flash device; boot flash device manufacturer part number; boot flash device catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	clock distribution device	clock distribution device; clock distribution device manufacturer part number; clock distribution device catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	high-current VRM module	high-current VRM module; high-current VRM module manufacturer part number; high-current VRM module catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	PCIe retimer set	PCIe retimer set; PCIe retimer set manufacturer part number; PCIe retimer set catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	power integrity capacitor bank	power integrity capacitor bank; power	no immediate evidence required	no	candidate remains out of

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
					integrity capacitor bank manufacturer part number; power integrity capacitor bank catalogue price GBP	unless this candidate is promoted to procurement scope		procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	BIOS firmware image	BIOS firmware image; BIOS firmware image manufacturer part number; BIOS firmware image catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	management Ethernet PHY	management Ethernet PHY; management Ethernet PHY manufacturer part number; management Ethernet PHY catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	platform configuration EEPROM	platform configuration EEPROM; platform configuration EEPROM manufacturer part number; platform configuration EEPROM catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	debug access lockout	debug access lockout; debug access lockout manufacturer	no immediate evidence required unless this candidate is promoted to	no	candidate remains out of procurement totals

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
					er part number; debug access lockout catalogue price GBP	procurement scope		

Procurement Readiness Gate

procurement_not_started

0/7 procurement areas pass; 2 review, 5 blocked. Critical production-eligible BoM rows 0/12; procurement-blocking closure rows 12; active sourcing rows 12; deferred sourcing rows 108. Source quality no_sourcing_evidence; evidence authenticity no_evidence; costing costing_not_started; BoM admission candidate_bom_authorized. Can use for procurement: no. Next action: Collect production-quality source evidence for critical BoM rows before procurement use..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture review	review	0.33	architecture_freeze_review_required; structurally ready yes, independent review accepted no.	none	Collect accepted engineering/interface/requirement review evidence before procurement use.
bom evidence trace	blocked	0	0/12 critical rows production-eligible; 12 critical unsourced, 0 protocol-only, 0 source-reference blocked.	structure_containment_rack_chassis_1 u_rack_chassis: Collect admissible source evidence for this critical line before BoM can pass. compute_acceleration_accelerator_module_ai_accelerator_module: Collect admissible source evidence for this critical line before BoM can pass. compute_acceleration_cpu_memory_complex_cpu_module: Collect admissible source evidence for this critical line before BoM can pass. compute_acceleration_cpu_memory_complex_ecc_memory : Collect admissible source evidence for this critical line before BoM can pass. compute_acceleration_cpu_memory_complex_system_motherboard: Collect admissible source evidence for this critical line before BoM can pass. plus 7 more	Collect production-quality source evidence for critical BoM rows before procurement use.
sourcing batch closure	blocked	0.9	12 procurement-blocking closure row(s); 12 active sourcing row(s), 108 deferred row(s).	bom-close:compute_acceleration_accelerator_module_ai_accelerator_module: Collect admissible source evidence for this critical line before BoM can pass. bom-close:compute_acceleration_cpu_memory_complex_cpu_module: Collect admissible source evidence for this	Work the active sourcing batches until procurement-blocking closure rows are closed.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				critical line before BoM can pass. bom-close:compute_acceleration_cpu_memory_complex_ecc_memory: Collect admissible source evidence for this critical line before BoM can pass. bom-close:compute_acceleration_cpu_memory_complex_system_motherboard: Collect admissible source evidence for this critical line before BoM can pass. bom-close:control_computer_communication_host_control_bmc_management_controller: Collect admissible source evidence for this critical line before BoM can pass. plus 7 more	
source reference quality	blocked	0	no_sourcing_evidence; 0/0 source-reference row(s) pass; 0 protocol, 0 placeholder.	none	Admit production source evidence before procurement readiness can start.
evidence authenticity	blocked	0	no_evidence; 0/0 evidence rows production-ready; 0 protocol.	No evidence records are present.	Add source and reviewer evidence before procurement readiness can start.
bom costing	blocked	0.14	costing_not_started ; critical priced 0/12; BoM total 0 GBP.	structure_containment_rack_chassis_1u_rack_chassis: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. compute_acceleration_accelerator_module_ai_accelerator_module: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. compute_acceleration_cpu_memory_complex_cpu_module: Collect admissible supplier/manufacturer/MPN/unit-cost	Close costing blockers before procurement use.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				evidence before BoM can pass. compute_accelerati on_cpu_memory_co mplex_ecc_memory : Collect admissible supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. compute_accelerati on_cpu_memory_co mplex_system_mot herboard: Collect admissible supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. plus 70 more	

prototype procurement policy	review	0	candidate_bom_aut horized; display mode candidate_only; prototype procurement use allowed no.	none	Prototype policy keeps procurement use disabled until external procurement review and approval are explicitly added.
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Sourcing Evidence Pack

Evidence packets request source data only. They do not admit supplier, manufacturer, MPN or cost claims.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
1U rack chassis	1 candidate	1U rack chassis; 1U rack chassis manufacturer part number; 1U rack chassis catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
AI accelerator module	1 candidate	AI accelerator module; AI accelerator module manufacturer part number; AI accelerator module catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
CPU module	1 candidate	CPU module; CPU module manufacturer part number; CPU module catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
ECC memory	1 candidate	ECC memory; ECC memory manufacturer part number; ECC memory catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
System motherboard	1 candidate	System motherboard; System motherboard manufacturer part number; System motherboard catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
High-speed NIC	1 candidate	High-speed NIC; High-speed NIC manufacturer part number; High-speed NIC catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
NVMe SSD	1 candidate	NVMe SSD; NVMe SSD manufacturer part number; NVMe SSD catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
BMC management controller	1 candidate	BMC management controller; BMC management controller manufacturer part number; BMC management controller catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
Redundant power supply	2 each	Redundant power supply; Redundant power supply manufacturer part number; Redundant power supply catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Fan wall assembly	1 candidate	Fan wall assembly; Fan wall assembly manufacturer part number; Fan wall assembly catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Heatsink cold plate	1 candidate	Heatsink cold plate; Heatsink cold plate manufacturer part number; Heatsink cold plate catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
TPM security module	1 candidate	TPM security module; TPM security module manufacturer part number; TPM security module catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Cost model status: pending sourcing. Architecture is being reviewed before cost claims are admitted.

Cost Bucket	Estimate
CAPEX	£0
Annual OPEX	£0
NRE	£25,000

No admitted supplier spend yet. Supplier rows appear only after source-backed evidence records are admitted.

Regulatory And Risk

Standard	Title	Jurisdiction	Evidence Required
IEC-62368-1	Audio/video, ICT and communication equipment safety	UK/EU/global	Electrical safety, enclosure, temperature and abnormal-operation evidence.
CE-EMC	Electromagnetic compatibility technical file	UK/EU	EMC design review, pre-compliance or test evidence for rack appliance.
SECURE-BOOT-REVIEW	Secure boot and fleet-management review	Project security gate	TPM, measured boot, firmware signing, recovery and management-access evidence.

Hazard	S	O	D	RPN	Mitigation
Accelerator thermal throttling or shutdown under rack inlet conditions	8	4	4	128	Thermal path sizing, inlet derating, fan redundancy and telemetry alarms.
Insecure firmware or management interface compromise	9	3	5	135	TPM measured boot, signed firmware, locked debug access and management-network controls.
Inference outage from PSU, fan or SSD failure	7	4	4	112	Redundant PSU path, hot-swap fans, storage health monitoring and runtime failover policy.

Section Issues And Score

bom

- blocker / critical_part_unpriced: 1U rack chassis is critical but has no credible unit cost.
- blocker / critical_part_unpriced: AI accelerator module is critical but has no credible unit cost.
- blocker / critical_part_unpriced: CPU module is critical but has no credible unit cost.
- blocker / critical_part_unpriced: ECC memory is critical but has no credible unit cost.
- blocker / critical_part_unpriced: System motherboard is critical but has no credible unit cost.
- blocker / critical_part_unpriced: High-speed NIC is critical but has no credible unit cost.
- blocker / critical_part_unpriced: NVMe SSD is critical but has no credible unit cost.
- blocker / critical_part_unpriced: BMC management controller is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Redundant power supply is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Fan wall assembly is critical but has no credible unit cost.

- blocker / critical_part_unpriced: Heatsink cold plate is critical but has no credible unit cost.
- blocker / critical_part_unpriced: TPM security module is critical but has no credible unit cost.
- major / low_priced_line_ratio: Only 0% of BoM lines have prices.

Section Score Ledger

Rule: start at 9, subtract 3 per blocker, 1.5 per major, 0.5 per minor, floor at 1. Mean score 7.67. This is a deterministic gate score from current validators, not an external reviewer score or proof of real-world engineering quality. Reviewer acceptance is tracked separately in the verification ledger.

[illegible]

Section	Base	B/M/m Issues	Deduction	Final	Target	Issue Deductions	Rationale
							is attached to this scored section.
sources_references	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
Mean	7.67						